

IT2011 – Artificial Intelligence and Machine Learning

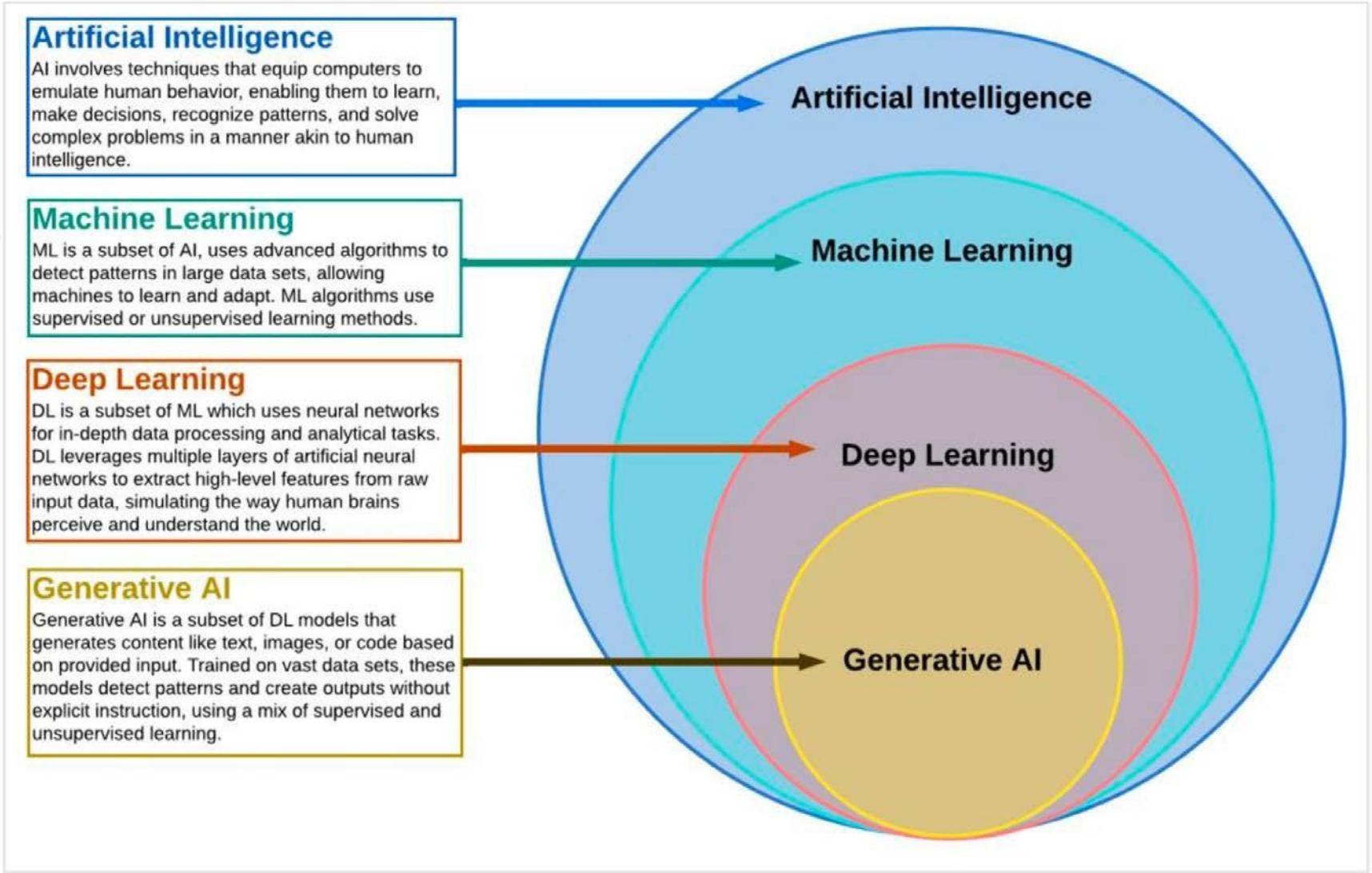
Lecture 05 - Introduction to Generative AI

July 2026

Prof. Nuwan Kodagoda

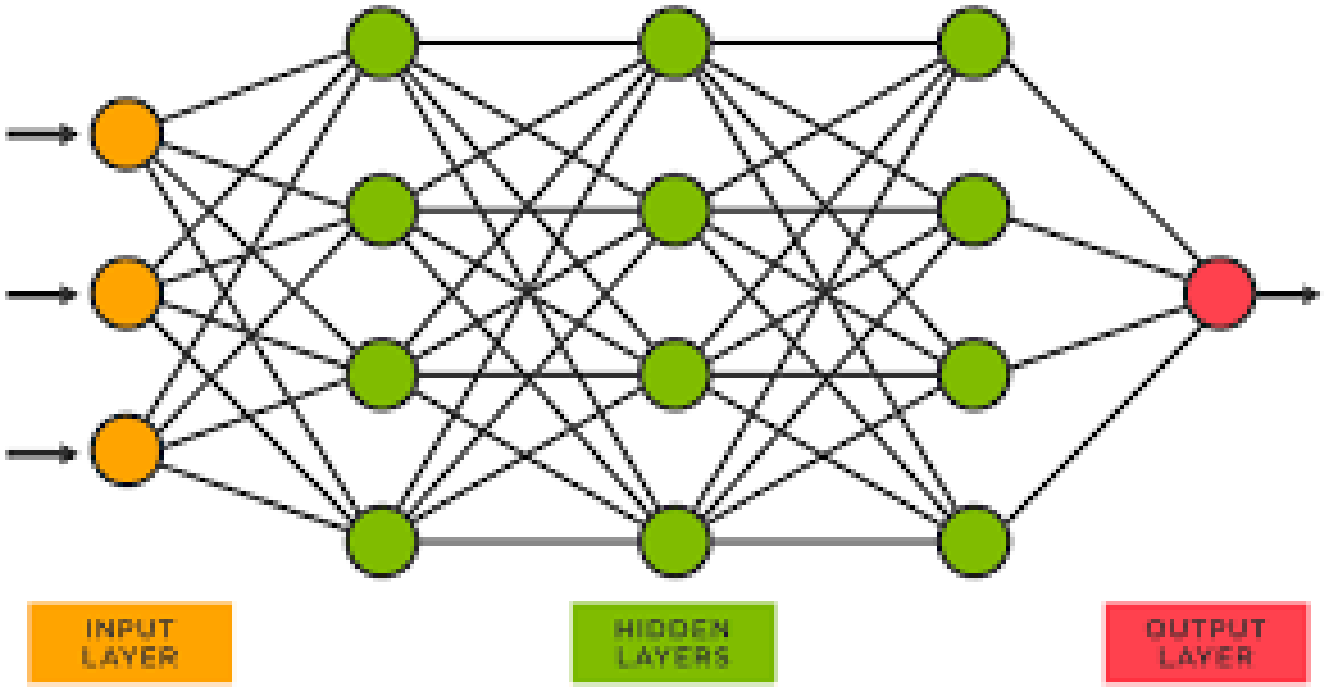
Agenda

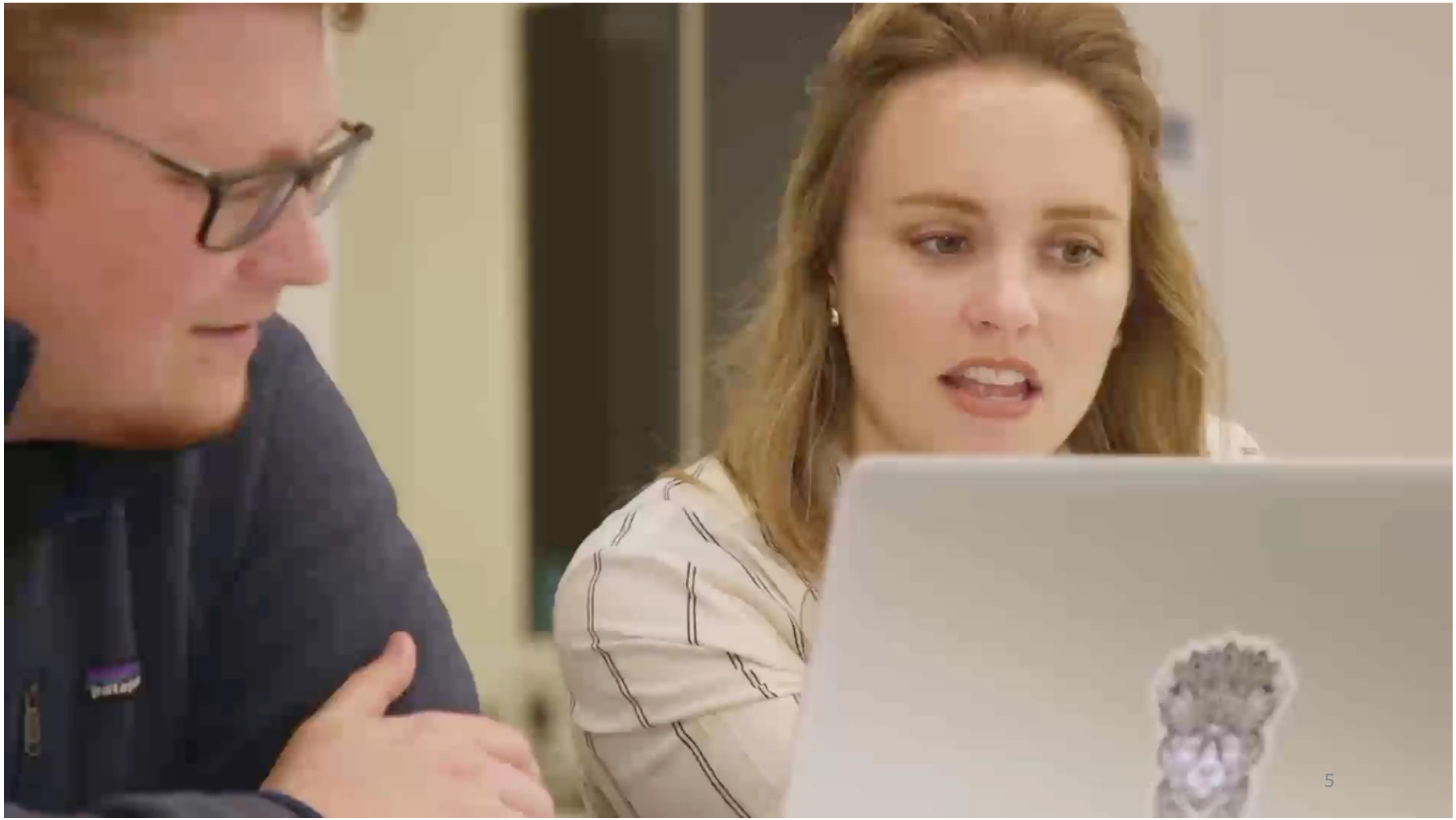
- Introductions
- Deep Neural Networks
- Generative AI Models
- Generative, Discriminative, Predictive AI
- GenAI for Education
- Developing GenAI based Applications
- GenAI definitions and Terms

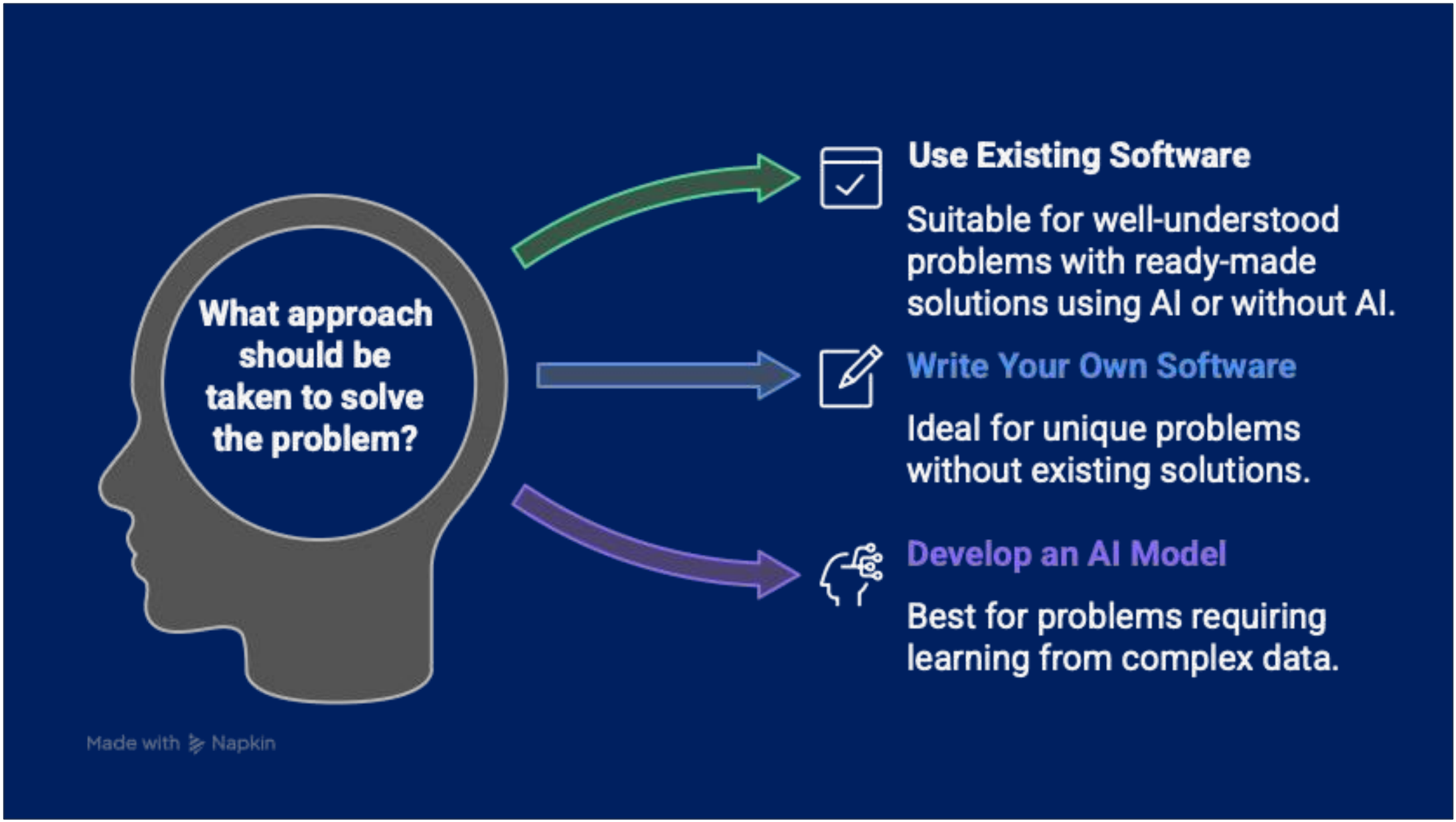


https://commons.wikimedia.org/wiki/File:Unraveling_AI_Complexity_-_A_Comparative_View_of_AI,_Machine_Learning,_Deep_Learning,_and_Generative_AI.jpg

A Neural Network

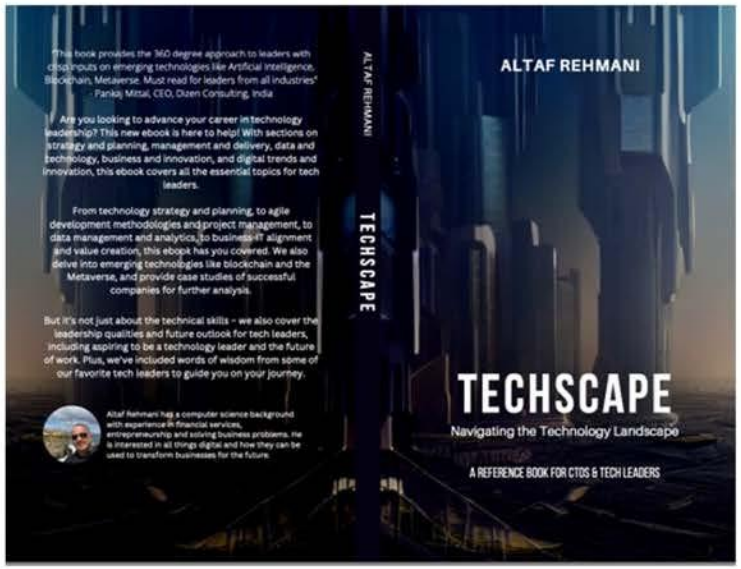





What approach should be taken to solve the problem?
Use Existing Software
Suitable for well-understood problems with ready-made solutions using AI or without AI.
Write Your Own Software
Ideal for unique problems without existing solutions.
Develop an AI Model
Best for problems requiring learning from complex data.
Made with Napkin

Generative AI fundametals

2022: Year of Generative AI: Text prompts generating cohesive text, Pictures, Video and a plethora of services helping us generate code, articles, websites, slides boosting productivity.



2022: Book For Technology Leaders Co-written with **chatGPT** in less than 10 hours



A female portrait closeup in the style of Van Goh: **Lexica.art**



A dog wearing a superhero cape flying through the sky using **Meta's Make-a-video**

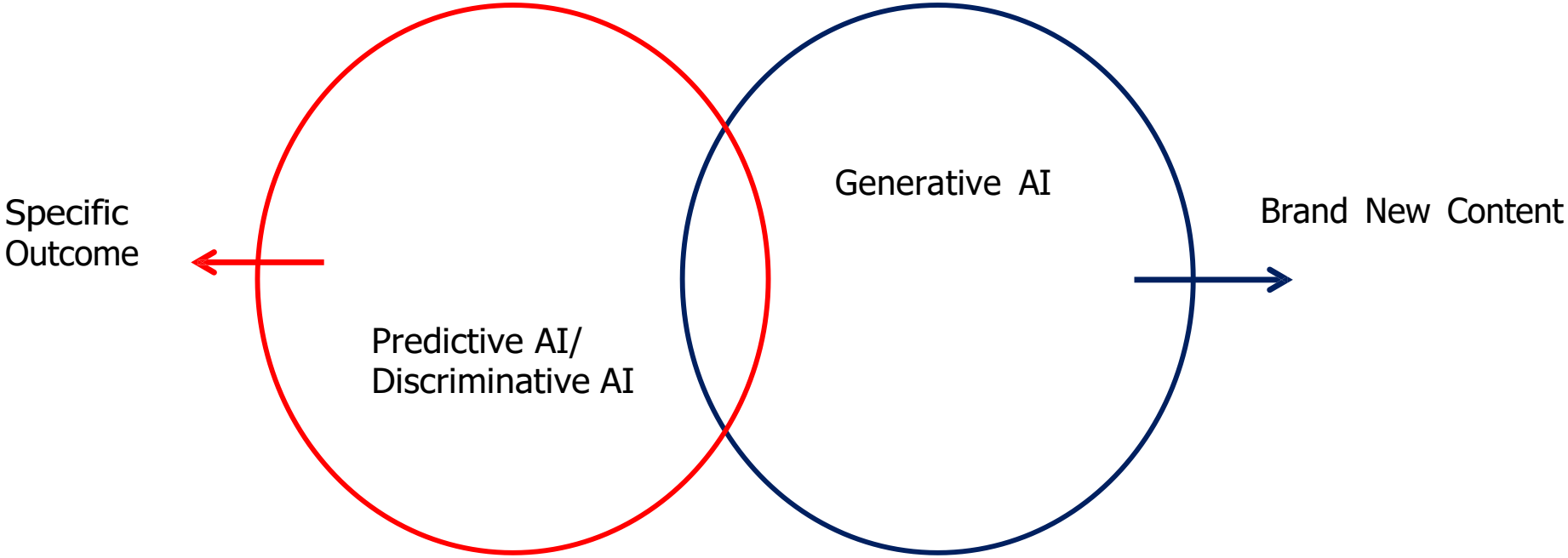
Reference - Altaf Amani, Generative AI for Everyone

What are the problems that are better suited for AI/ML applications?

a) Calculating grades in a School Examination
b) Predicting next weeks Vegitable prices
c) Converting Hand written text to printed text
d) Calculating the Salary of an Employee
e) Launching Rockets
f) Checking Grammer
g) Generating Music, Art

Introductions

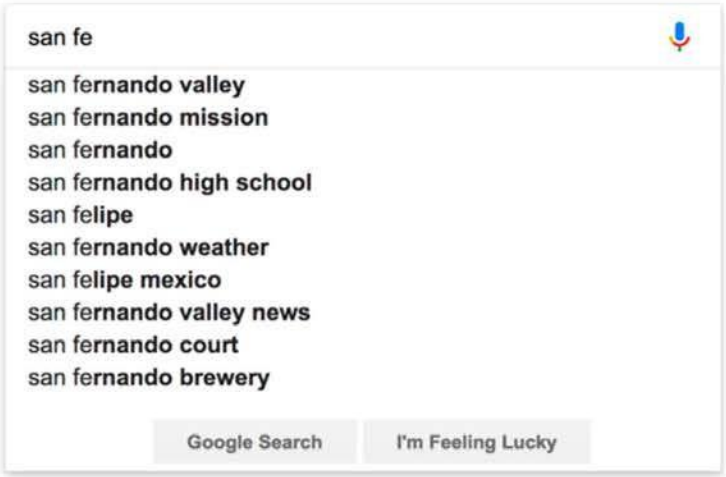
Generative vs Predictive AI

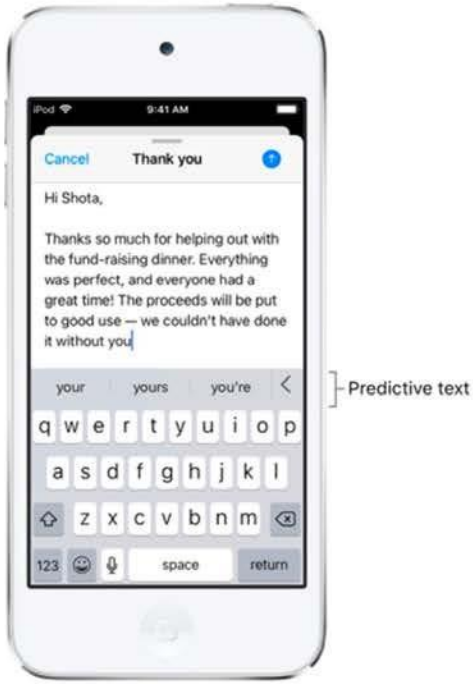


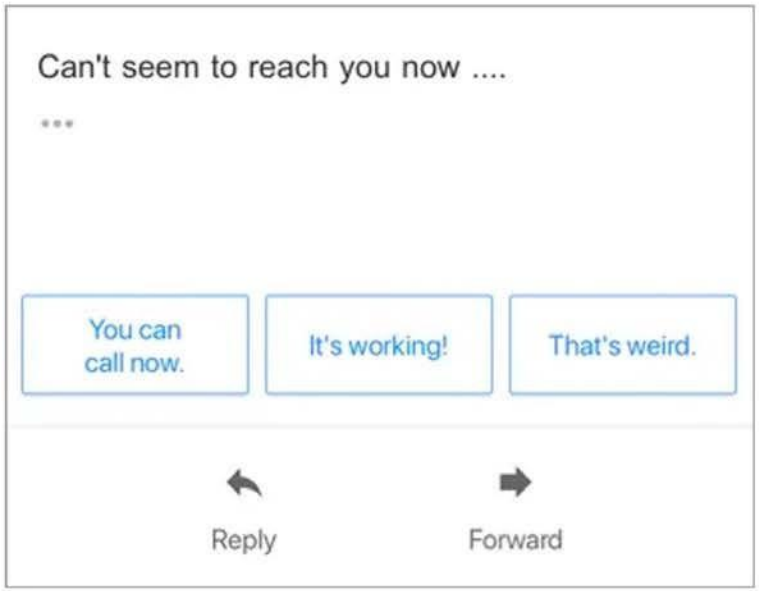
Reference - Altaf Amani, Generative AI for Everyone

Feature	Discriminative AI	Predictive AI	Generative AI
Goal	Classifies existing data into categories	Predicts future values based on past data	Generates entirely new data
Output Type	Discrete labels (e.g., cat vs. dog)	Continuous or categorical values (e.g., stock price)	New content (e.g., images, text, videos)
Example Tasks	Spam detection, facial recognition, fraud detection	Weather forecasting, sales prediction, churn prediction	Text generation, image synthesis, music composition
Common Models	Logistic regression, SVMs, neural networks for classification	Regression models, LSTMs, time-series forecasting	GANs, VAEs, Transformer models (GPT, DALL·E)
Example Models	Random forests, CNNs, decision trees	ARIMA, LSTMs, deep learning for forecasting	GPT-4, Stable Diffusion, Midjourney, StyleGAN

What is a Language Model

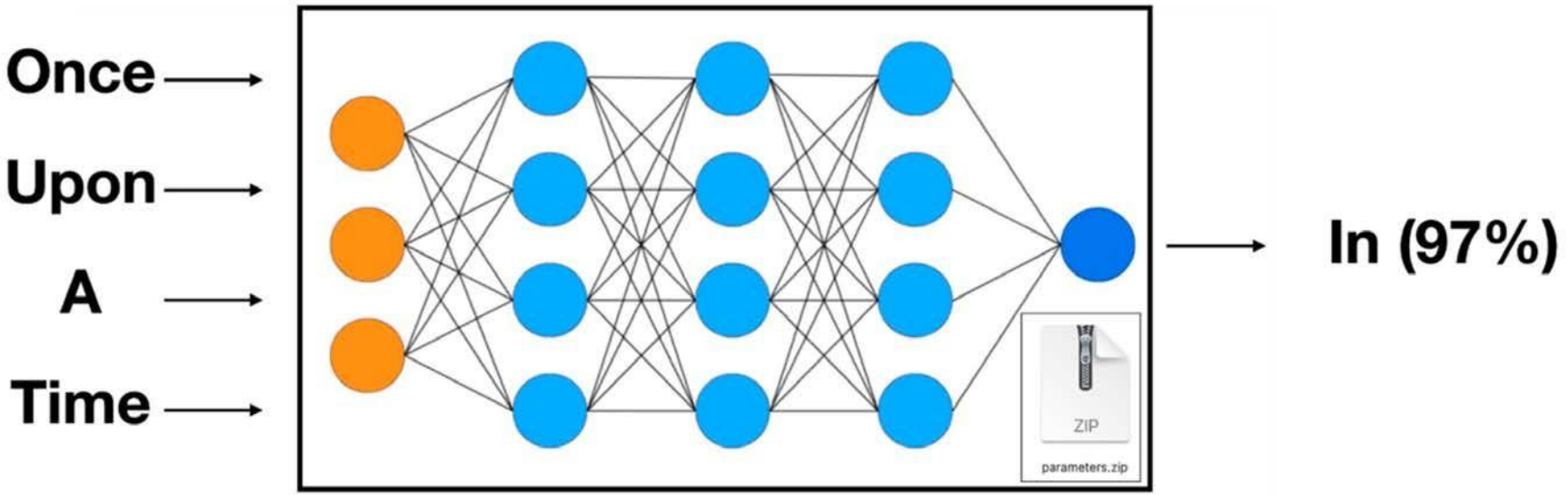






Reference - Altaf Amani, Generative AI for Everyone

Neural Networks for Language Models



e.g. context of 4 words

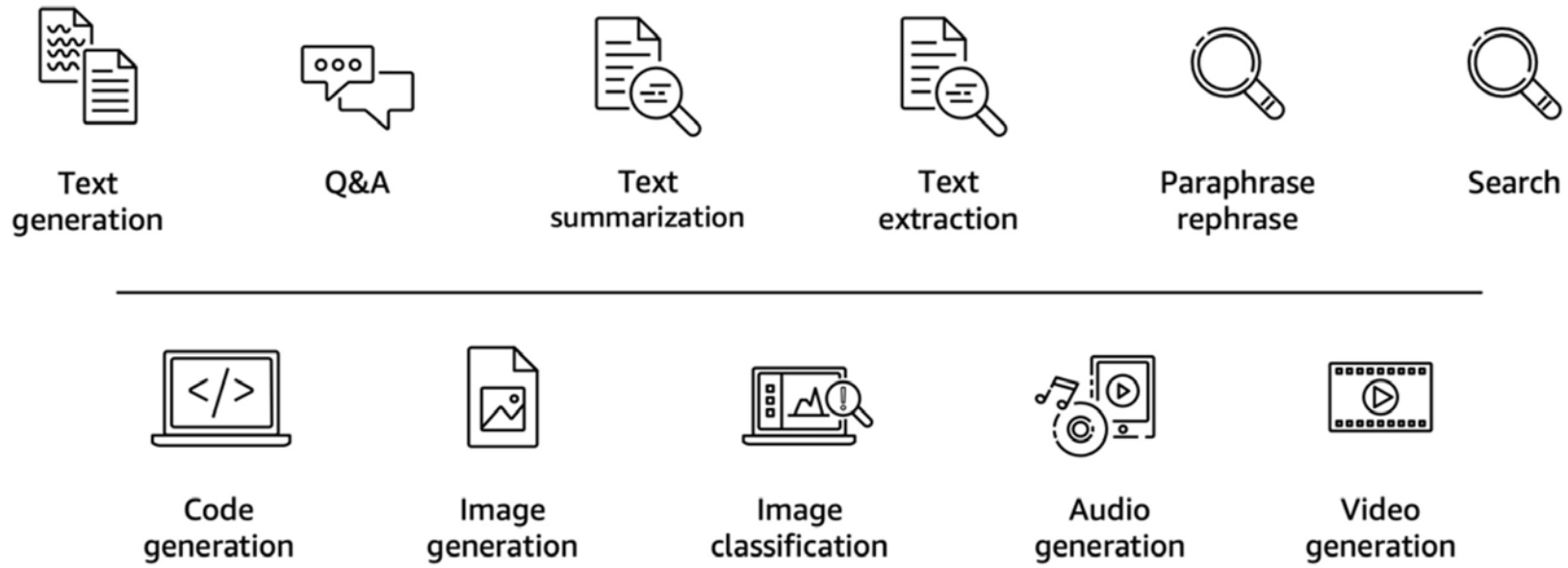
Auto regresive Models – ChatGPT
Auto encoding Models - Gemini

predict next word

Modern LLMs use the Transformer Architecture introduced in 2017, by Google in the Research Paper "Attention is all you need"

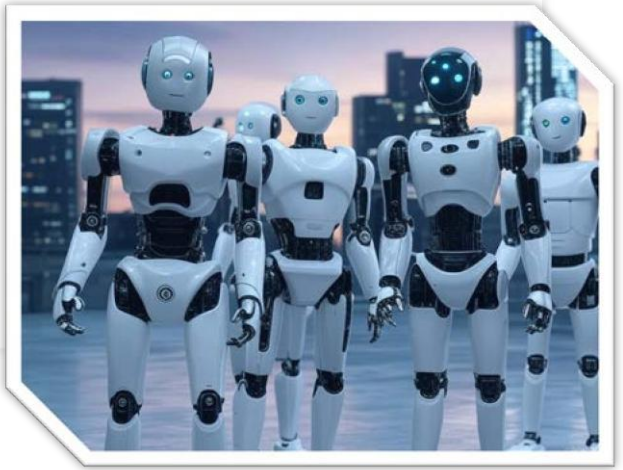
Reference - Altaf Amani, Generative AI for Everyone

Use cases of Generative AI



Reference - Altaf Amani, Generative AI for Everyone

News Articles about DeepSeek R1, Qwen 2.5 Max, ChatGPT, Grok 3



1

A shocking Chinese AI advancement called DeepSeek is sending US stocks plunging



By David Goldman and Matt Egan, CNN

5 minute read · Updated 4:21 PM EST, Mon January 27, 2025



2

Home » Feature Story » General » Qwen 2.5 Max better than DeepSeek, beats ChatGPT in coding, costs 10x less than Claude 3.5

Qwen 2.5 Max better than DeepSeek, beats ChatGPT in coding, costs 10x less than Claude 3.5

By Sagar Sharma | Updated on 31-Jan-2025

4

OpenAI

OpenAI launches 'deep research' tool that it says can match research analyst

ChatGPT developer announces AI agent amid growing challenge from rivals such as China's DeepSeek

Dan Milmo *Global technology editor*

Mon 3 Feb 2025 13.26 GMT

Share



3

OpenAI Fights Back Against DeepSeek AI With Early o3-Mini Launch—Here's How It Compares

The release of OpenAI's o3-mini comes as DeepSeek's ultra-efficient R1 model sparked a trillion-dollar tech selloff and reshaped the AI landscape.



By Jose Antonio Lanz

Feb 1, 2025

6 min read

5

Business / Tech

Elon Musk debuts Grok 3, an AI model that he says outperforms ChatGPT and DeepSeek



By Clare Duffy, CNN

3 minute read · Updated 1:39 PM EST, Tue February 18, 2025


Grok


deepseek

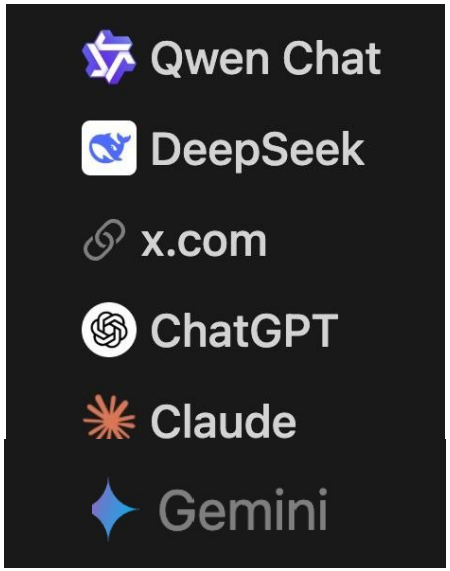

ChatGPT


Claude


Gemini


Qwen

Reasoning Models



- Reasoning models, like **OpenAI** *o1*, *o3-mini-high*, and *Deep Research*, **DeepSeek** *DeepThink R1*, **Gemini** 1.5 *Deep Research*, **Grok** 3 *Deep Search,* Qwen 2.5 Max *Thinking (QwQ)* are large language models trained with reinforcement learning to perform complex reasoning.
- Reasoning models think before they answer, producing a long internal chain of thought before responding to the user. Reasoning models excel in complex problem solving, coding, scientific reasoning, and multi-step planning for agentic workflows.

LLM Demos

GPT 4.5 has passed turing test ?

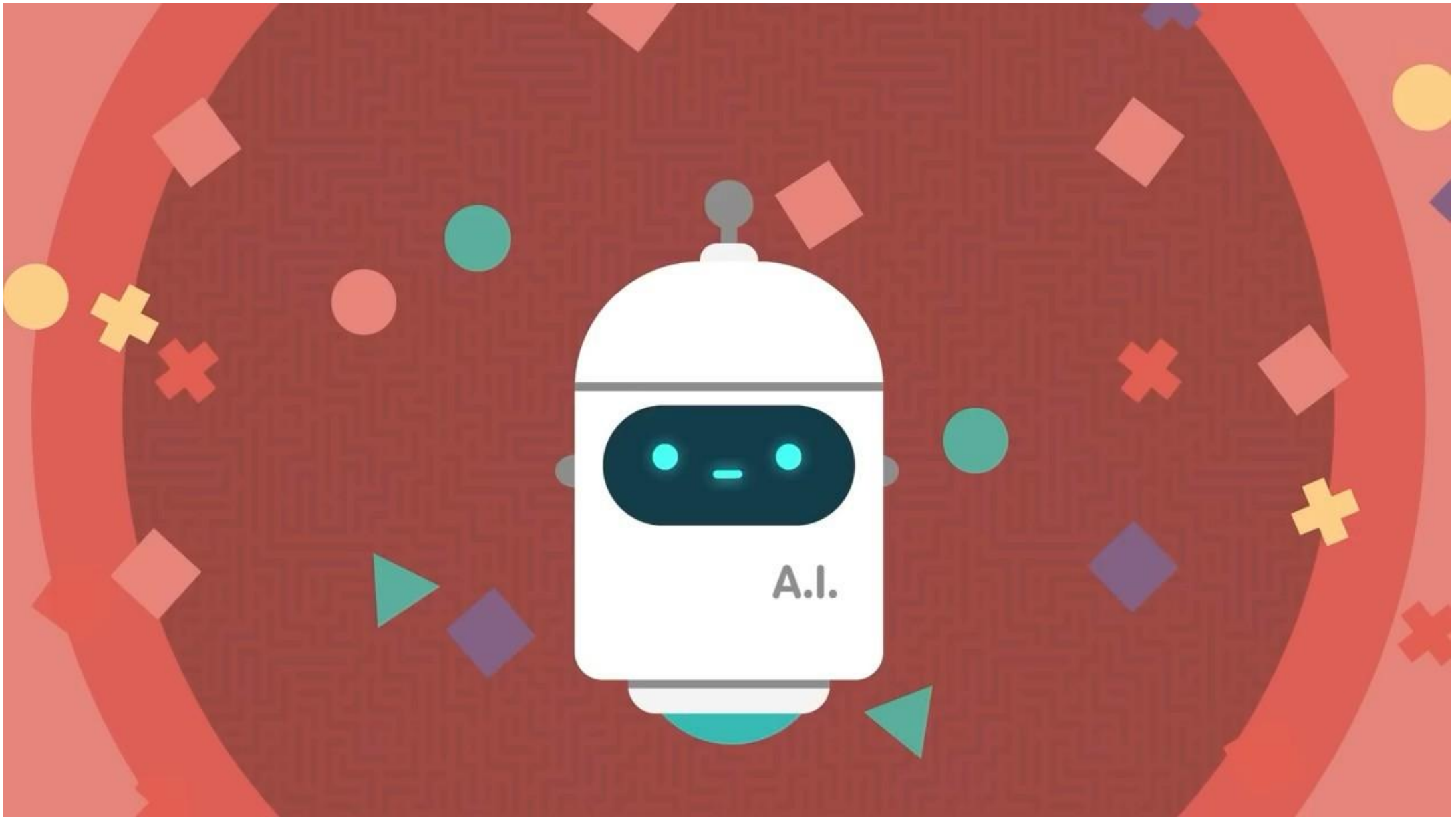
[Submitted on 31 Mar 2025]

Large Language Models Pass the Turing Test

Cameron R. Jones, Benjamin K. Bergen

We evaluated 4 systems (ELIZA, GPT-4o, LLaMa-3.1-405B, and GPT-4.5) in two randomised, controlled, and pre-registered Turing tests on independent populations. Participants had 5 minute conversations simultaneously with another human participant and one of these systems before judging which conversational partner they thought was human. When prompted to adopt a humanlike persona, GPT-4.5 was judged to be the human 73% of the time: significantly more often than interrogators selected the real human participant. LLaMa-3.1, with the same prompt, was judged to be the human 56% of the time -- not significantly more or less often than the humans they were being compared to -- while baseline models (ELIZA and GPT-4o) achieved win rates significantly below chance (23% and 21% respectively). The results constitute the first empirical evidence that any artificial system passes a standard three-party Turing test. The results have implications for debates about what kind of intelligence is exhibited by Large Language Models (LLMs), and the social and economic impacts these systems are likely to have.

Agentic Demo


A.I.

Prompt Engineering

Effective Prompt Generation

1. Context
2. Task
3. Instructions
4. Clarify and Refine

You are an educational expert who specializes in <Field>, teaching <level> students following an undergraduate degree in <field>. Create a lesson plan that is for 60 minute on the <topic>. The lesson plan must cater to higher order thinking skills must encapsulate UDL principles <etc.>.

Do you understand?

Prompt Generation

"Prompting" refers to the input or question that you provide to the model. The model takes this prompt and generates a response based on the information it has been trained on.

Define the Objective	• Identify the specific information or assistance
Be Clear and Precise	• Use clear language and avoid ambiguity. • Include essential details without overcomplicating the prompt.
Consider Context	• Provide relevant background or context to guide the response.
Set the Tone and Style	• Specify the desired tone (formal, casual) or style (e.g., summary, explanation) if it matters for your use case.
Ask Direct Questions	• If seeking specific information, formulate your prompt as a direct question. • Self-Reflective
Avoid Bias and Leading Questions	• Craft the prompt neutrally to prevent biased or skewed responses.
Test and Refine	• Experiment with different phrasings and observe how slight changes can affect the response. • Refine the prompt
Consider Ethical and Privacy Concerns	• Ethical guidelines and does not request or reveal sensitive or private information.

Prompt Generation

Aspect	Bad Prompt Example	Good Prompt Example
Clarity	"Write something about tech and security."	"Generate a report on emerging trends in cybersecurity technologies for IoT devices from 2022-2024."
Specificity	"Find articles on engineering stuff."	"Create a detailed overview of recent advancements in biomaterials for prosthetics, focusing on studies published in the last 3 years."
Feasibility	"Predict the future stock prices of major tech companies."	"Simulate financial scenarios for a medium-sized tech company considering a shift to cloud-based accounting, using current market data."
Ethical Boundaries	"Use confidential financial reports to analyze a company's internal investment strategies."	"Analyze public domain datasets to identify trends in investment patterns in the renewable energy sector."
Research Orientation	"Tell me if AI is better than humans in programming"	"Examine the impact of artificial intelligence on workflow efficiency in software development projects, referencing studies from 2020-2023."

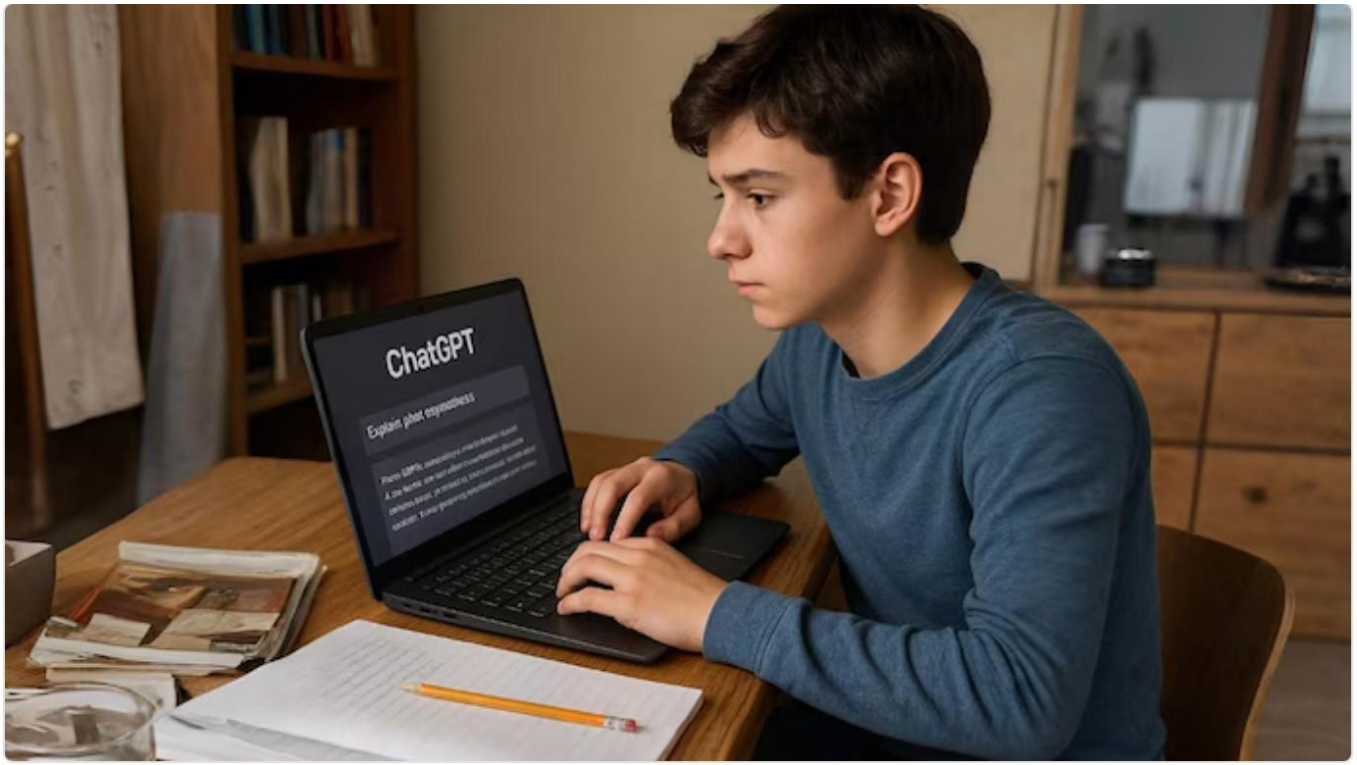
Coursera and AWS Courses

Using GenAI as a Learning Tool

- Using ChatBots using a Socrative Approach
- Your Personalized AI Assistant
- Brainstorming Ideas
- Generating FAQs, summaries of subject content
- Helping you to prepare for Exams, by generating questions and allowing you to answer them.

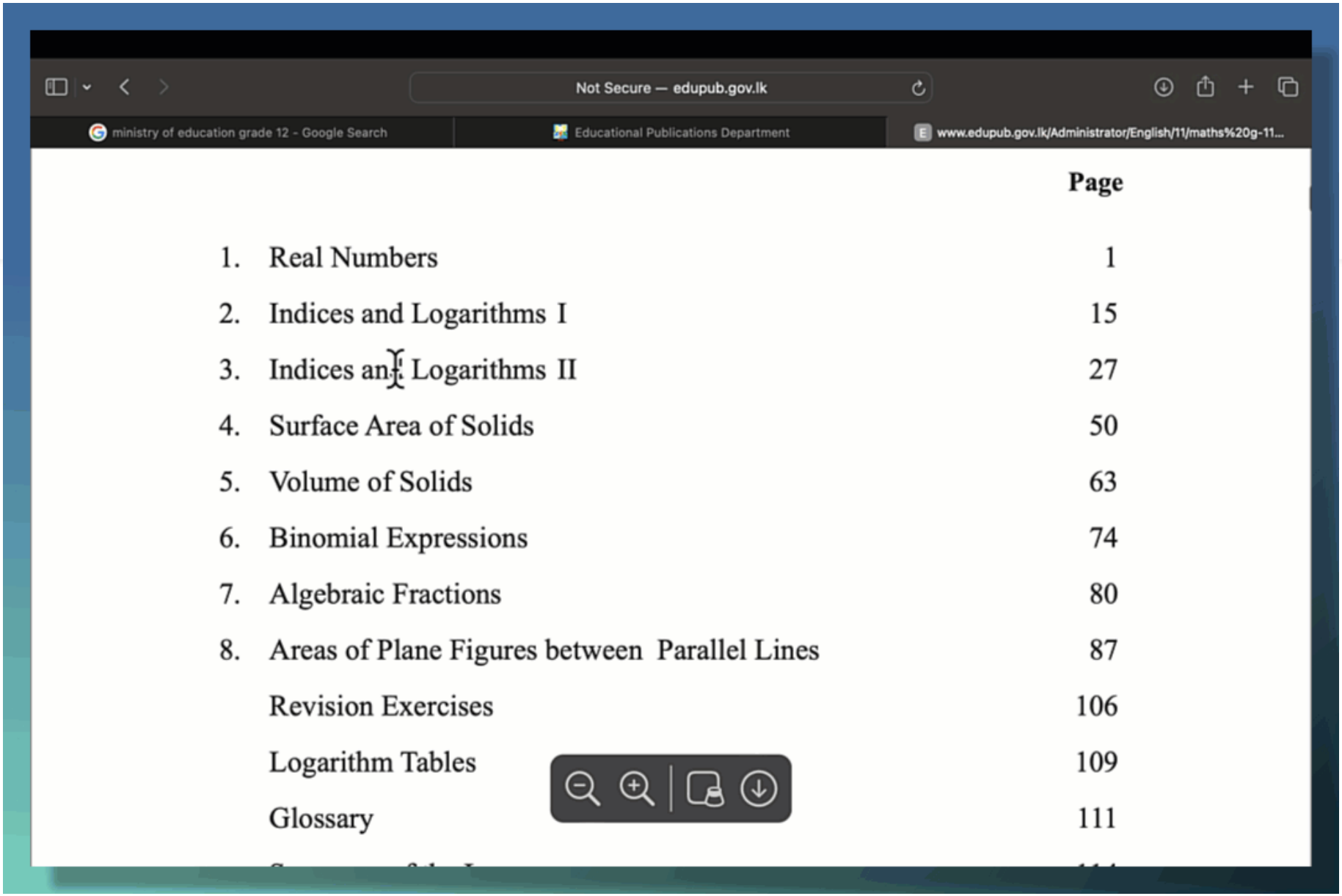
Is ChatGPT making us dumb? MIT study says students are using their brains less

A new study suggests that AI tools like ChatGPT can gradually impact our ability to learn and quickly analyse information. This is because they essentially allow humans to do certain tasks without using a lot of their brain power or thinking.

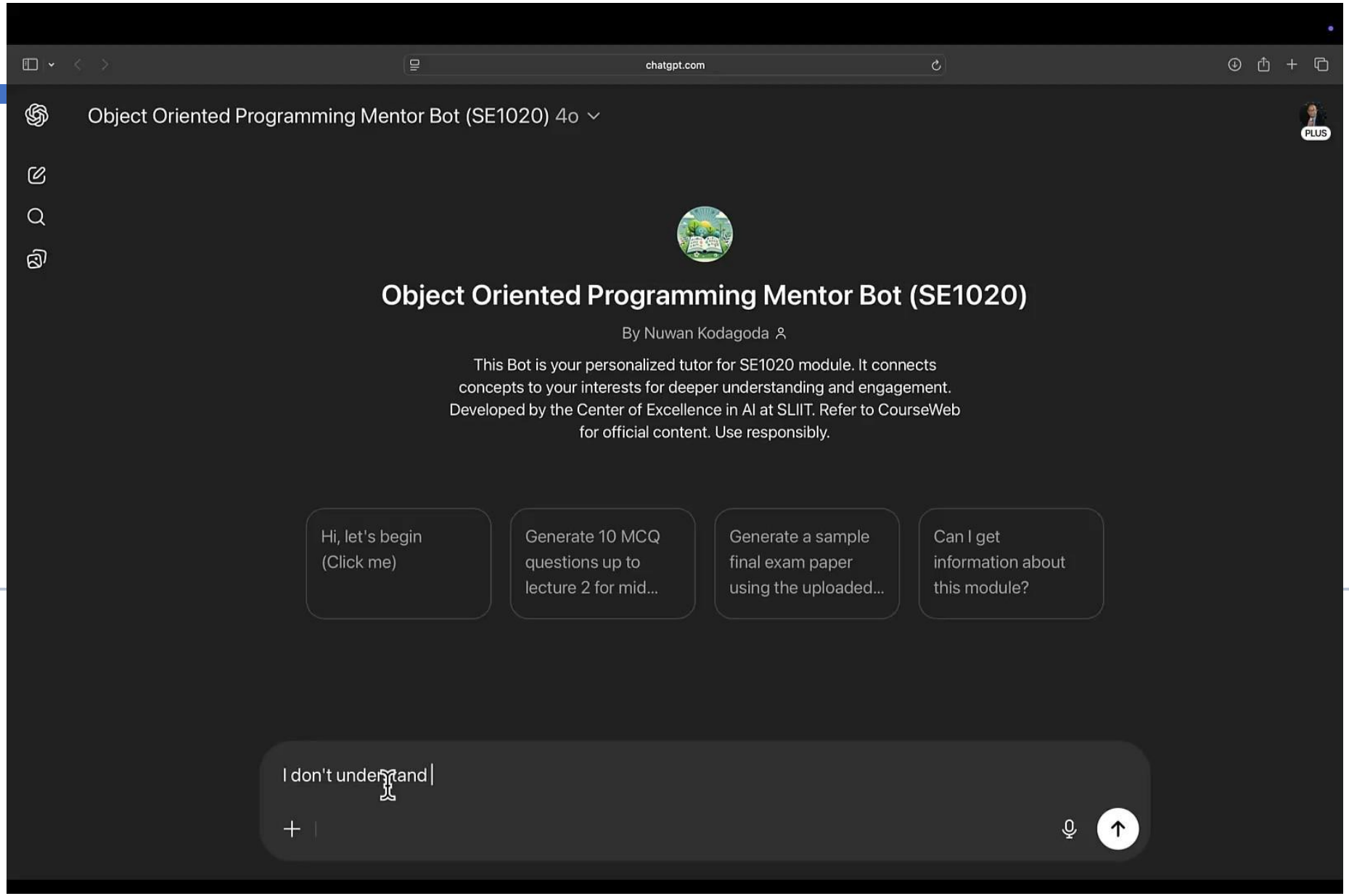


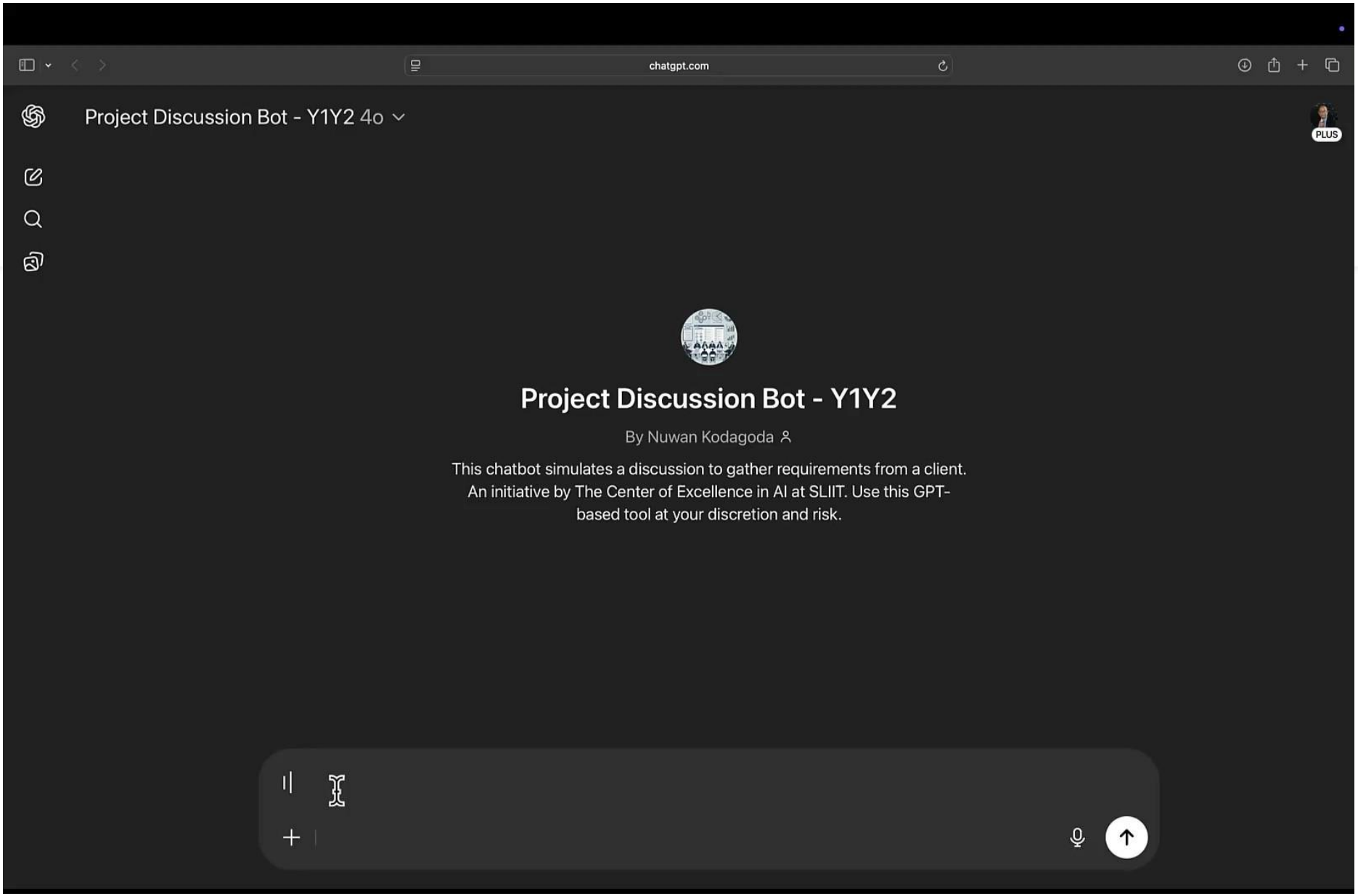
Why You Shouldn't Use ChatGPT just to Get Answers

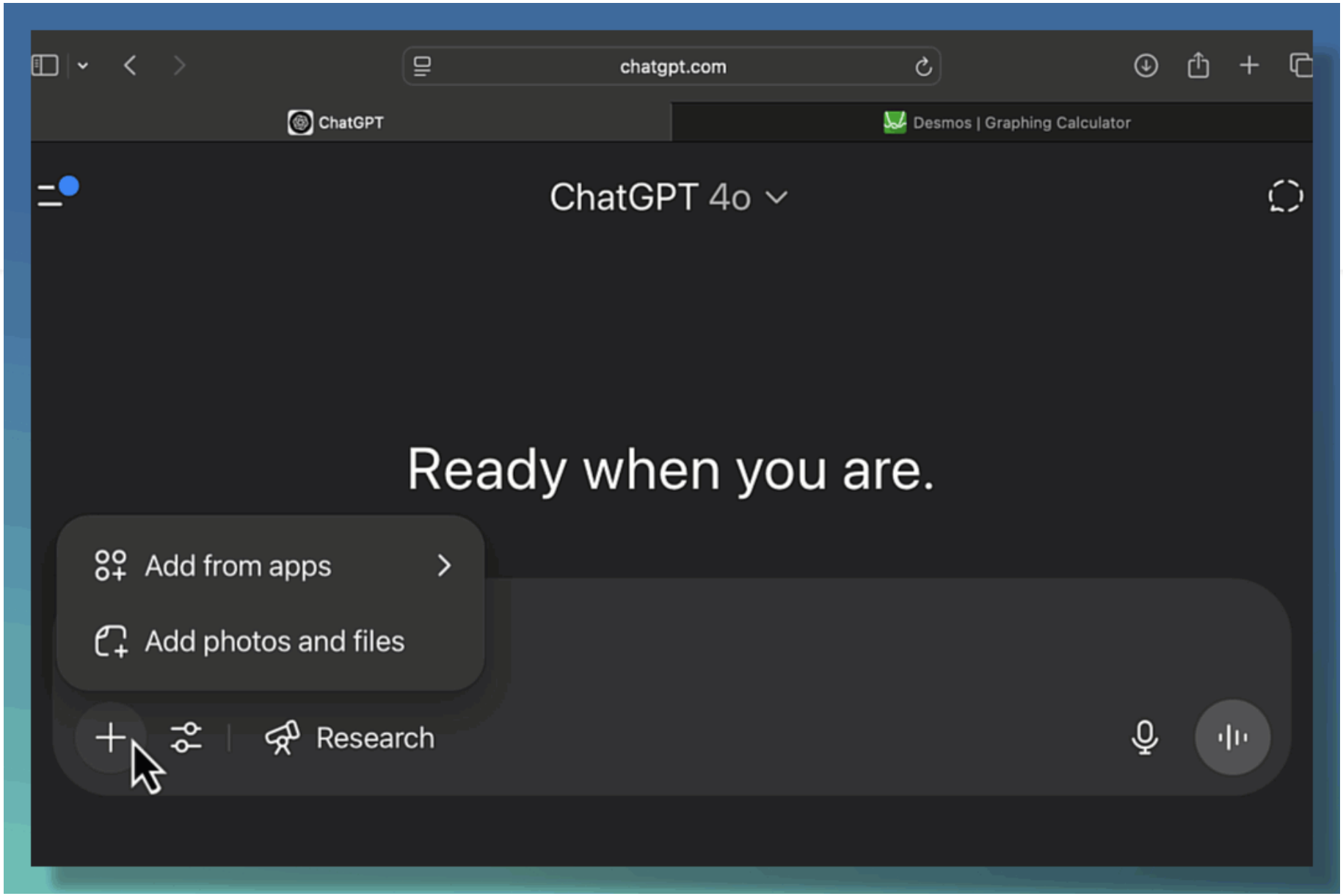
- Your brain learns best when it works hard.
- Using ChatGPT for answers makes your brain less active.
- Brain activity drops when you don’t think for yourself.
- Thinking builds memory and real understanding.
- Copying from AI = less learning happens.
- Use AI to guide you —— not to do the work for you.
- University is about learning how to think, not just finding answers.



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Object Oriented Programming Mentor Bot (SE1020) 4o
PLUS
Object Oriented Programming Mentor Bot (SE1020)
By Nuwan Kodagoda
This Bot is your personalized tutor for SE1020 module. It connects concepts to your interests for deeper understanding and engagement. Developed by the Center of Excellence in AI at SLIIT. Refer to CourseWeb for official content. Use responsibly.
Hi, let's begin (Click me)
Generate 10 MCQ questions up to lecture 2 for mid...
Generate a sample final exam paper using the uploaded...
Can I get information about this module?
I don't understand


chatgpt.com
Project Discussion Bot - Y1Y2 4o
PLUS
Project Discussion Bot - Y1Y2
By Nuwan Kodagoda
This chatbot simulates a discussion to gather requirements from a client. An initiative by The Center of Excellence in AI at SLIIT. Use this GPT-based tool at your discretion and risk.


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ChatGPT
Desmos | Graphing Calculator
ChatGPT 4o
Ready when you are.
Add from apps
Add photos and files
Research

UDACITY
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Home Discover ▾ Catalog

Help ▾ NK

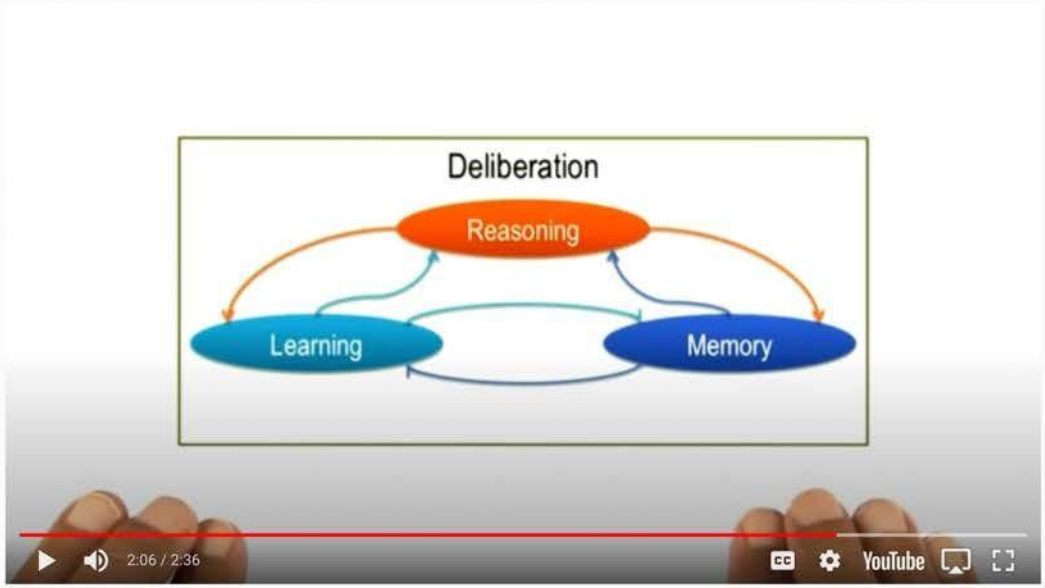
Knowledge-Based AI: Cognitive Systems

2%

01 - Intro to Knowledge-Based AI

- 1.1 Introductions
- 1.2 Preview
- 1.3 Conundrums in AI
- 1.4 Characteristics of AI Problems
- 1.5 Characteristics of AI Agents
- 1.6 Exercise: What are AI Problems?
- 1.7 Exercise: AI in Practice: Watson
- 1.8 What is Knowledge-Based AI?
- 1.9 Foundations: The Four Schools of AI
- 1.10 Exercise: What is KBAI?

Downloads



+ Show Video Transcript

Optional Reading: "What is AI anyway?" - Ashok K. Goel, Jim Davies

Click here to view the full Knowledge-Based AI course.

← Previous Next →

Give Page Feedback

Udacity AI
Powered by OpenAI

further questions or need clarification on specific aspects, feel free to ask!

how does this relate to think fast and slow

The concepts of "thinking fast" and "thinking slow" relate to the different modes of cognitive processing described by psychologist Daniel Kahneman in his book "Thinking, Fast and Slow." Here's how these ideas connect to prompting and reasoning in AI:

1. **Thinking Fast (System 1)**: This mode of thinking is characterized by quick, automatic, and often intuitive responses. It relies on

Show Prompts +

Ask Udacity AI anything about this program

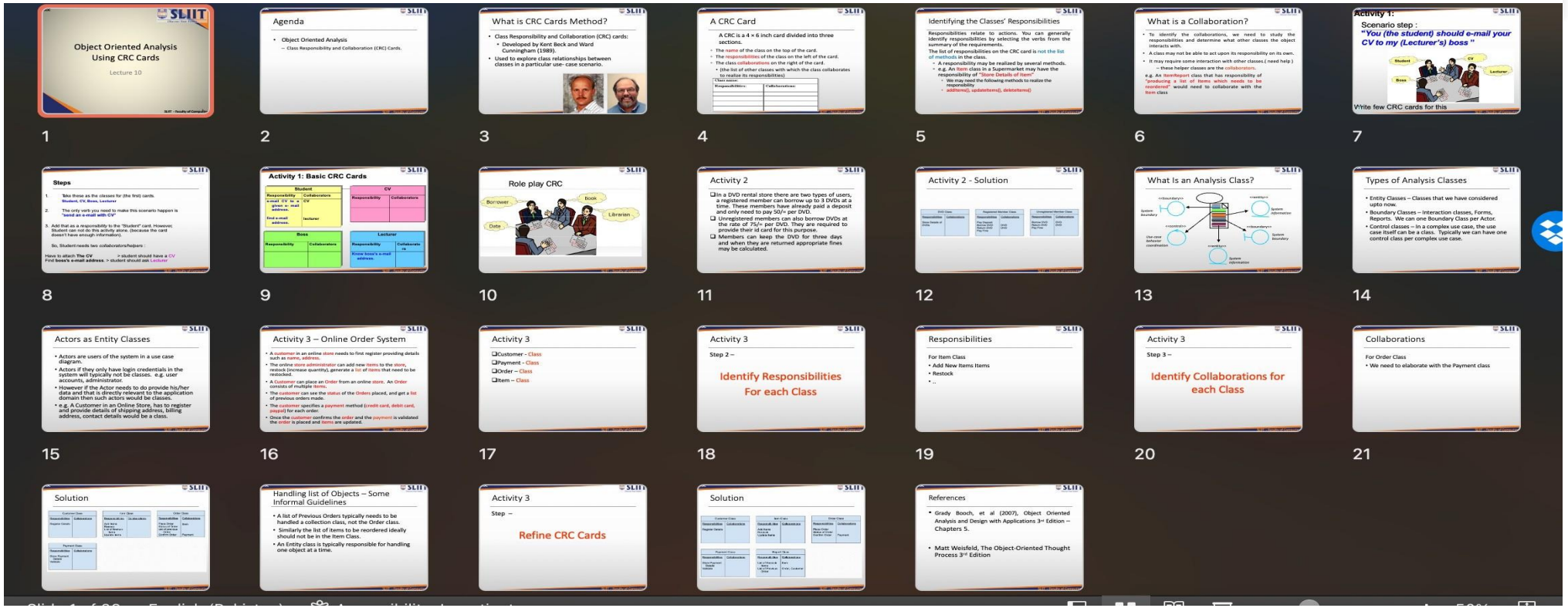
Send

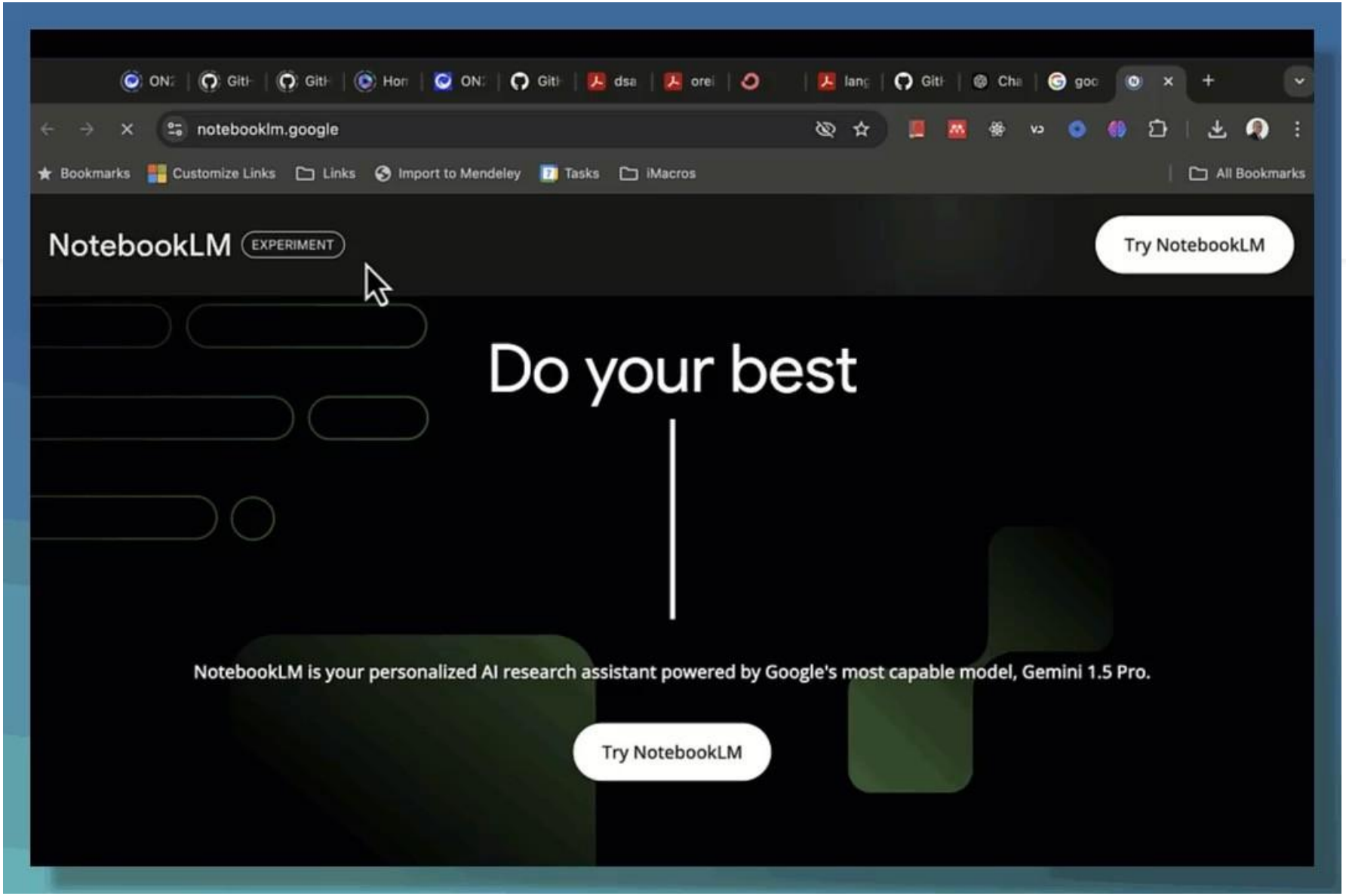
Need to speak to a mentor? Go to Knowledge

Need technical support? Go to Support

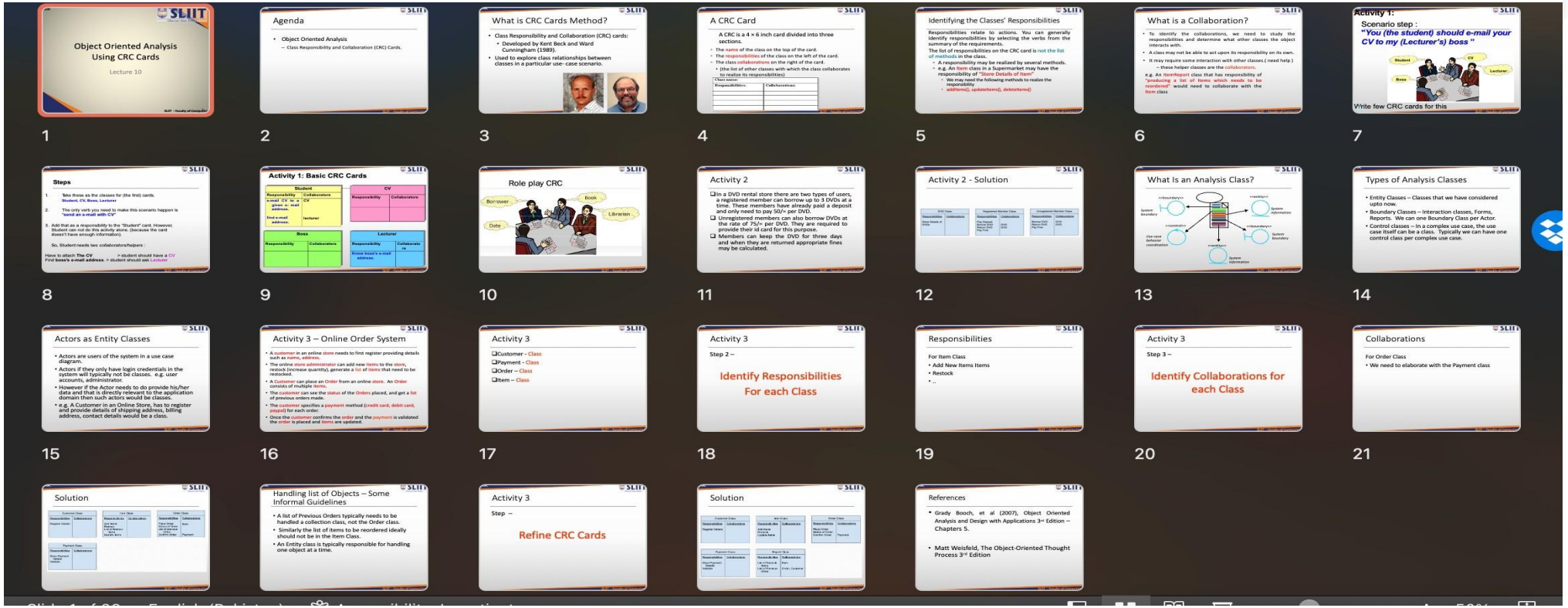
Power Point Coach

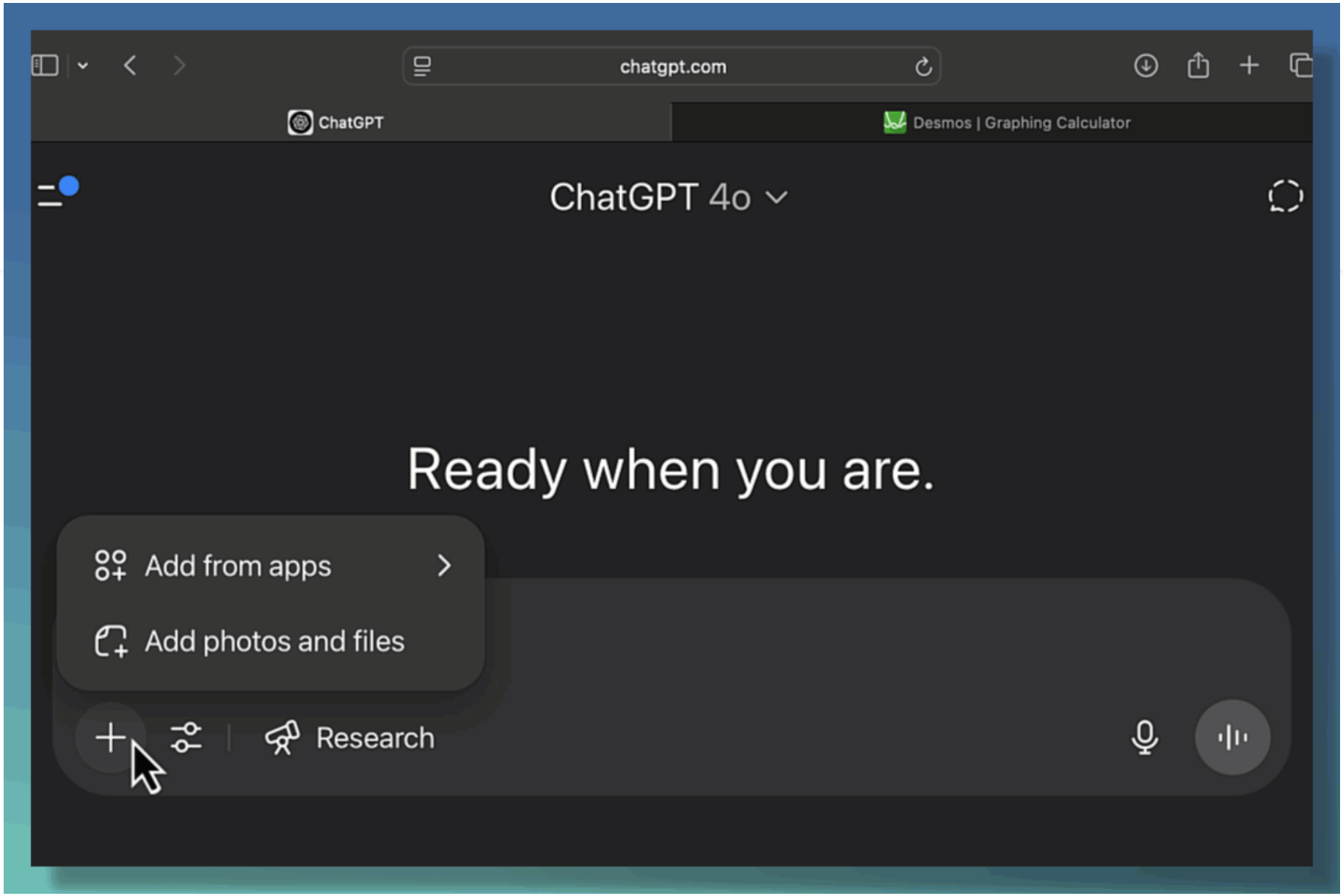
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Links
Import to Mendeley
Tasks
iMacros
All Bookmarks
NotebookLM
EXPERIMENT
Try NotebookLM
Do your best
NotebookLM is your personalized AI research assistant powered by Google's most capable model, Gemini 1.5 Pro.
Try NotebookLM

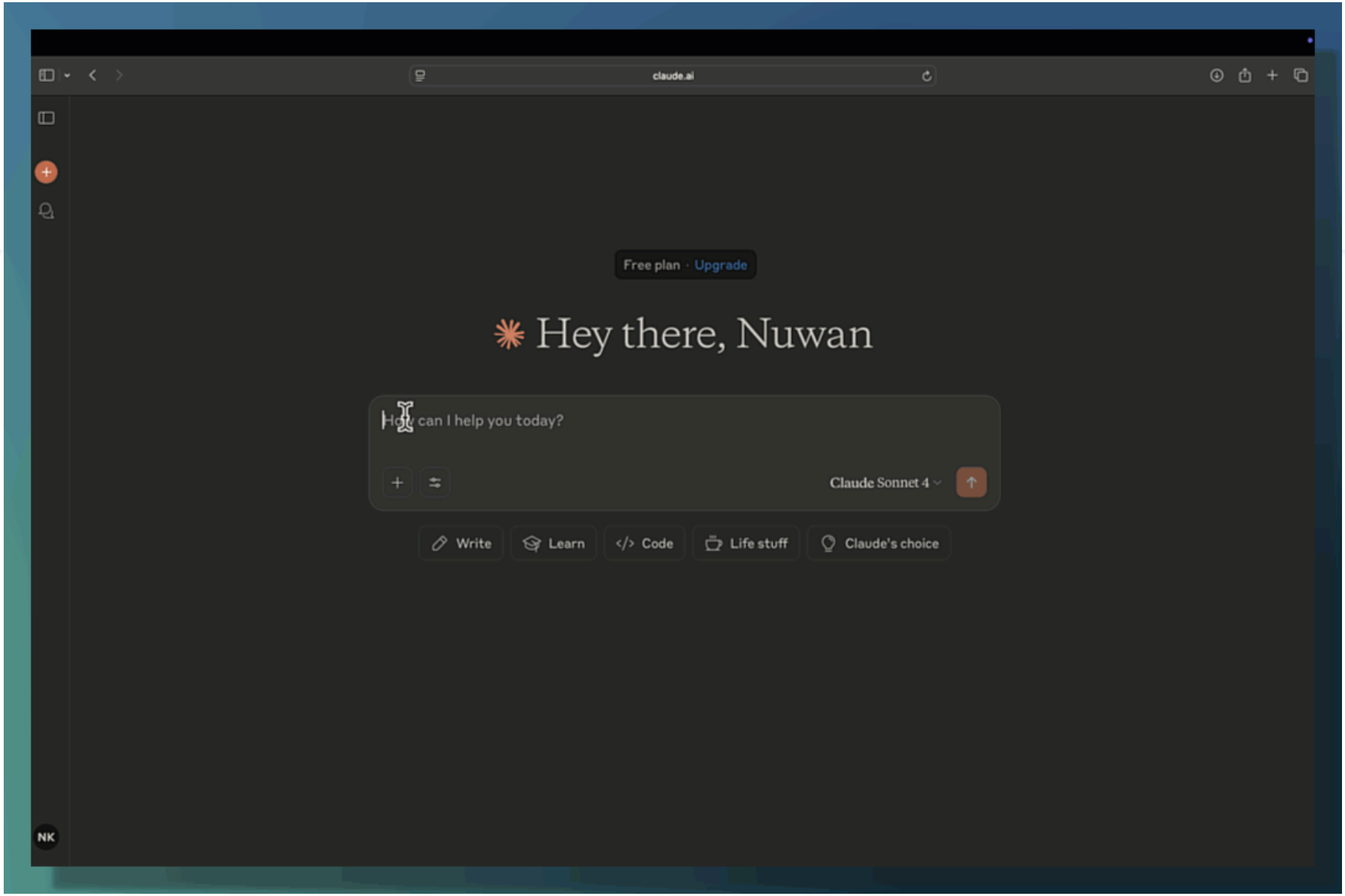
How about a podcast?



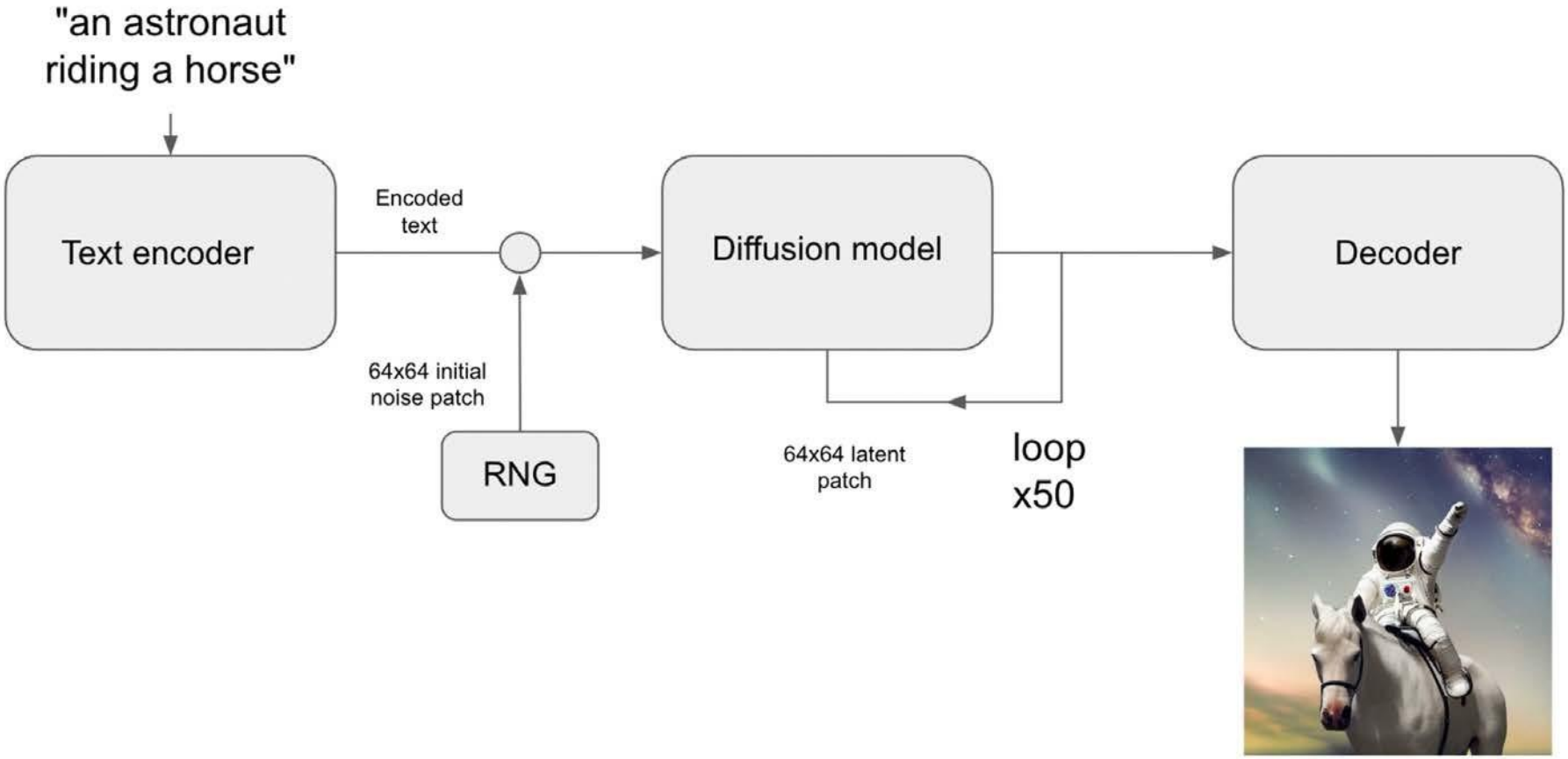

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ChatGPT
Desmos | Graphing Calculator
ChatGPT 4o
Ready when you are.
Add from apps
Add photos and files
Research

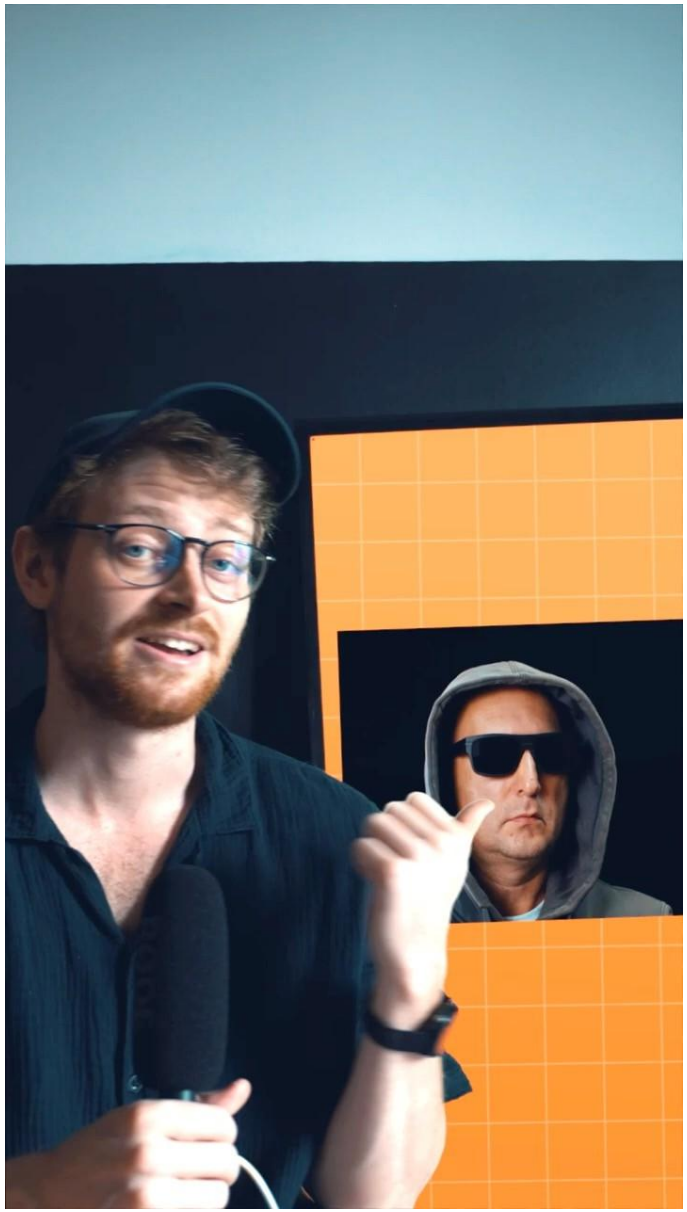
Using Claude for learning visual content

- I want to learn how to draw the graph $y = 2x + 5$

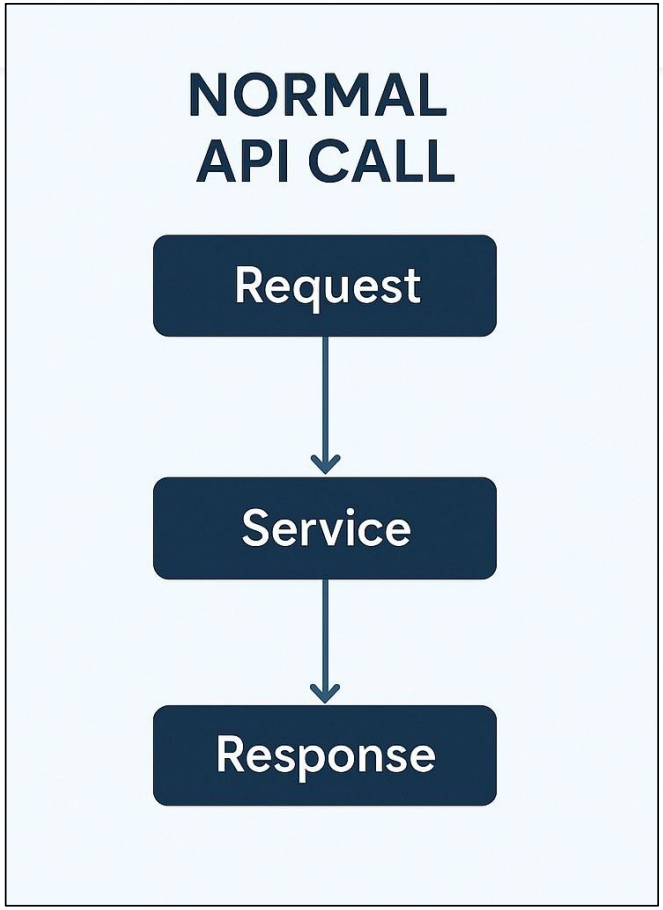

claude.ai
Free plan · Upgrade
Hey there, Nuwan
How can I help you today?
Claude Sonnet 4
Write
Learn
Code
Life stuff
Claude's choice
NK

Diffusion Models





Normal API Call in developing a software




```
import requests

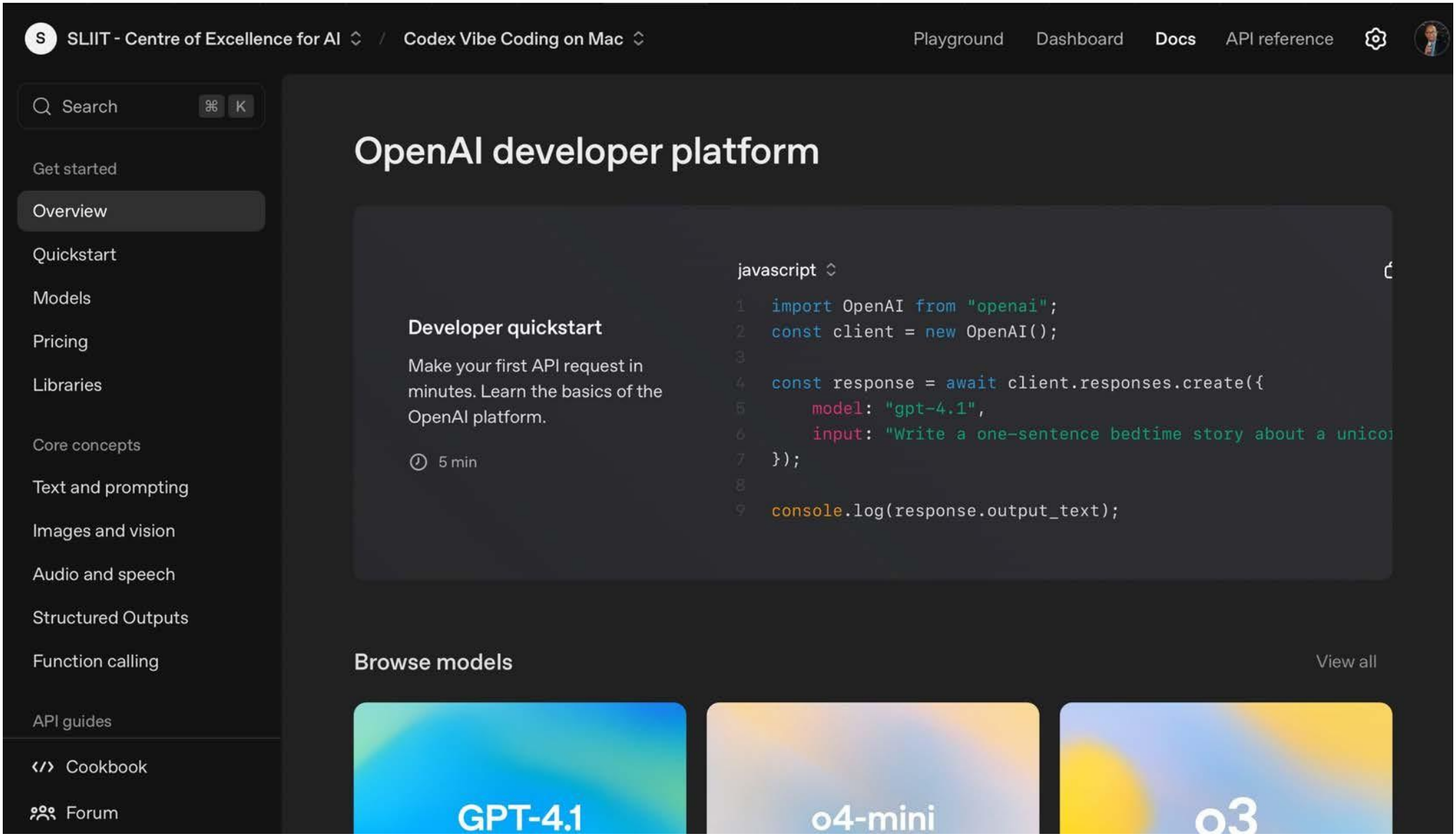
# Normal API call to a weather service
url = "https://api.weatherapi.com/v1/current.json"
params = {
    "key": "your_api_key",
    "q": "Colombo"
}

response = requests.get(url, params=params)
data = response.json()

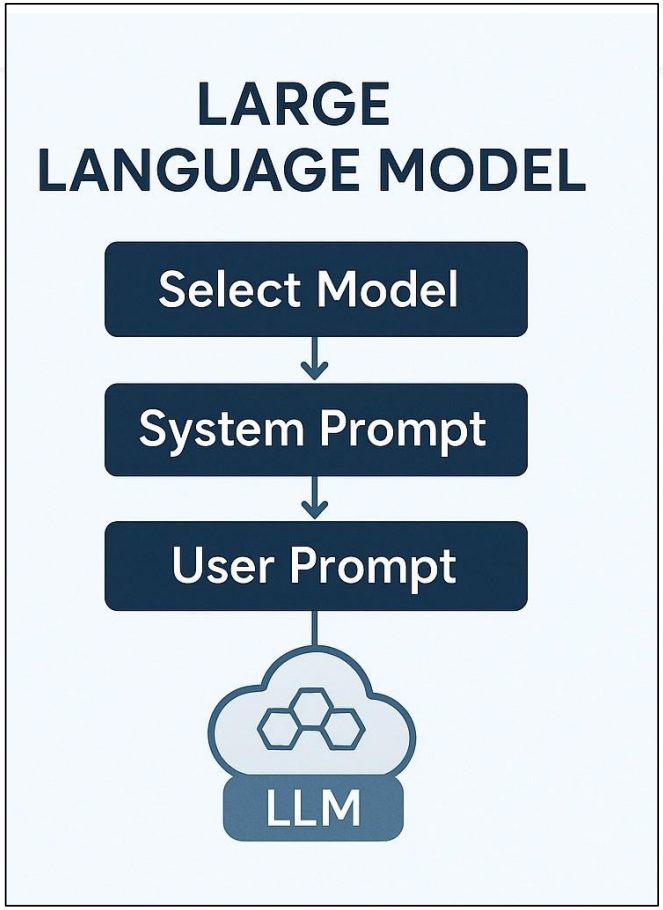
# Output part
print("Weather in Colombo:",
data["current"]["condition"]["text"])
print("Temperature:", data["current"]["temp_c"], "°C")
```

```
Weather in Colombo: Partly cloudy
Temperature: 30.0 °C
```

Developing GenAI Applications


SLIIT - Centre of Excellence for AI
Codex Vibe Coding on Mac
Playground
Dashboard
Docs
API reference
Search
⌘ K
Get started
Overview
Quickstart
Models
Pricing
Libraries
Core concepts
Text and prompting
Images and vision
Audio and speech
Structured Outputs
Function calling
API guides
Cookbook
Forum
OpenAI developer platform
Developer quickstart
Make your first API request in minutes. Learn the basics of the OpenAI platform.
5 min
javascript
import OpenAI from "openai";
const client = new OpenAI();
const response = await client.responses.create({
model: "gpt-4.1",
input: "Write a one-sentence bedtime story about a unico
});
console.log(response.output_text);
Browse models
View all
GPT-4.1
o4-mini
o3

How we call an LLM to develop our own Application



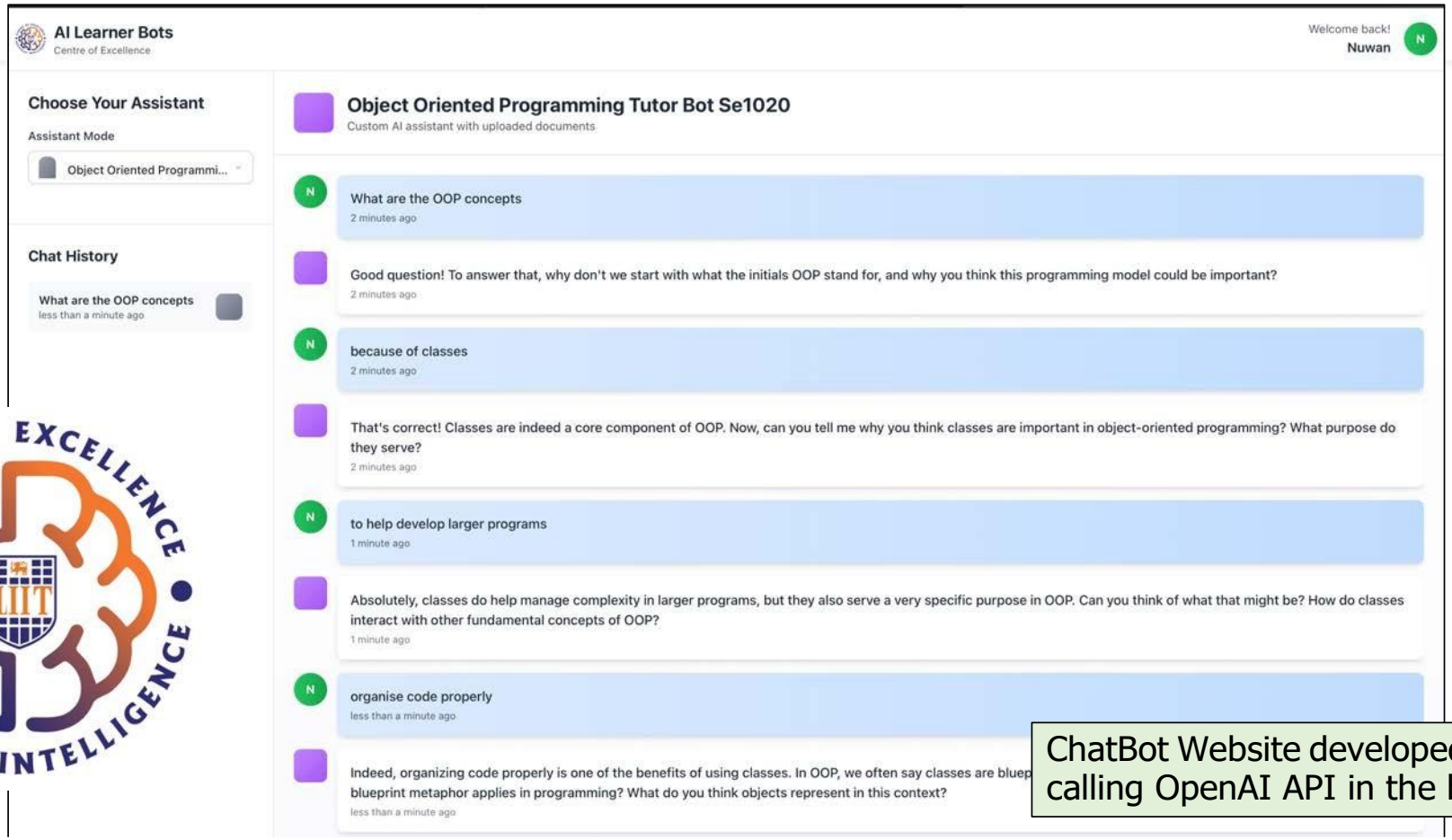

```
import openai

openai.api_key = "your_api_key"

response = openai.ChatCompletion.create(
    model="gpt-4",
    messages=[
        {"role": "system", "content": "You are a helpful
                assistant that explains weather reports."},
        {"role": "user", "content": "What does it mean when
                the weather report says 'partly cloudy with
                a chance of showers'?"}
    ]
)
print(response['choices'][0]['message']['content'])
```

```
"Partly cloudy with a chance of showers" means that there
will be a mix of sun and clouds throughout the day, and
there's a possibility of some rain occurring, but it's not
guaranteed. You might experience dry periods with occasional
light rain or brief showers.
```

Newer Versions of Learning Bots under development by FOC



ChatBot Website developed using React calling OpenAI API in the backend

Open and Closed Models

- Open Models:
  - Publicly available weights and code
  - Community-driven improvement
  - Customizable and transparent
  - Examples: LLaMA, Mistral, Falcon, GPT-J

- Closed Models:
  - Proprietary and controlled access
  - Trained on large private datasets
  - Examples: GPT-4 / GPT-4o, Claude 2/3, Gemini, Grok (xAI)

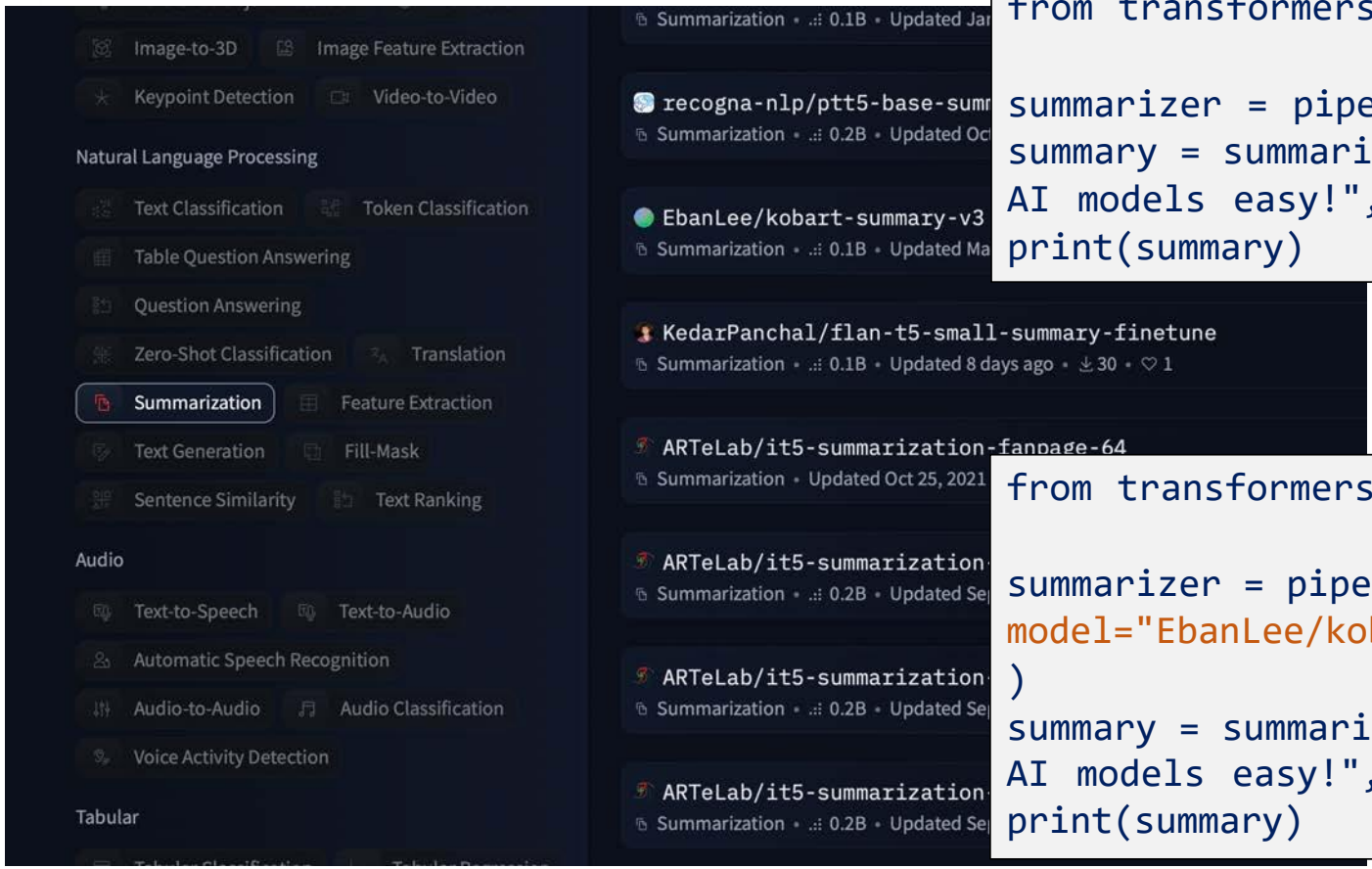
Hugging Face

- An open-source platform and company focused on AI/NLP.
- Hosts a hub of pre-trained models and datasets.
- Offers tools to work with text, images, audio, and more.

🤗 Transformers Library – Access to top AI models.
📦 Model Hub – Thousands of models for direct use.
🧪 Datasets – Ready-to-use datasets for ML tasks.
🌐 Inference API – Run models instantly in the cloud.
🔧 Spaces – Host ML apps using Gradio or Streamlit.

- No need to train models from scratch.
- Time-saving with pre-built models.
- Large, active community.
- Supports PyTorch, TensorFlow, JAX.
- Integration-friendly with real-world apps.

Loading and using a Hugging Face Model



General Call

```
from transformers import pipeline

summarizer = pipeline("summarization")
summary = summarizer("Hugging Face makes using
AI models easy!", max_length=20)
print(summary)
```

Specific Model Call

```
from transformers import pipeline

summarizer = pipeline("summarization”,
model="EbanLee/kobart-summary-v3"
)
summary = summarizer("Hugging Face makes using
AI models easy!", max_length=20)
print(summary)
```

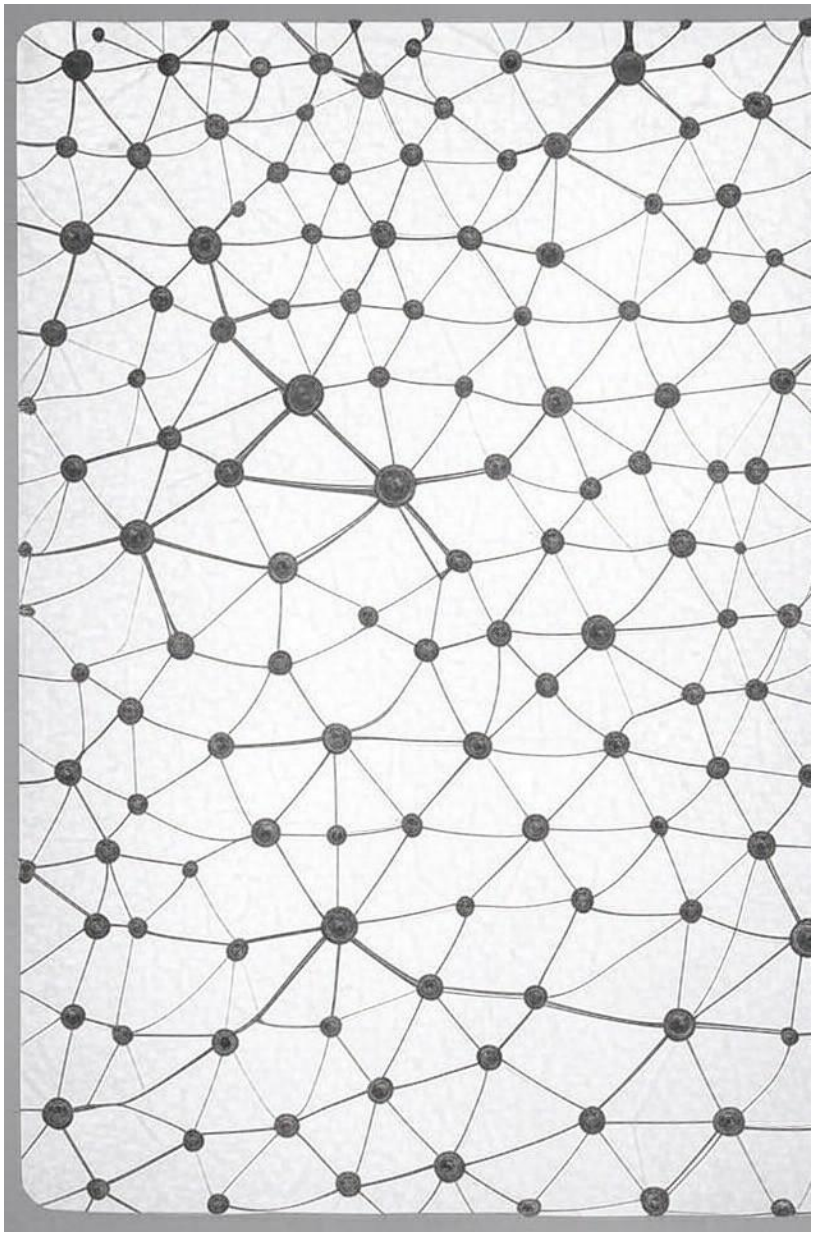
Loading and using a Hugging Face Model

```
from transformers import pipeline

summarizer = pipeline("summarization")
summary = summarizer("""Hugging Face is a pioneering open-source AI platform
  that empowers developers, researchers, and organizations to easily access and
  deploy cutting-edge machine learning models for natural language processing, computer vision,
  and beyond-democratizing AI and accelerating innovation through a collaborative ecosystem of tools,
  datasets, and community-driven contributions.""", max_length=20)
print(summary)
```

```
No model was supplied, defaulted to sshleifer/distilbart-cnn-12-6 and revision a4f8f3e (https://huggingface.co/sshleifer/distilbart-cnn-12-6).
Using a pipeline without specifying a model name and revision in production is not recommended.
config.json: 1.80k/? [00:00<00:00, 94.2kB/s]
pytorch_model.bin: 100% 1.22G/1.22G [00:07<00:00, 269MB/s]
model.safetensors: 94% 1.14G/1.22G [00:05<00:00, 197MB/s]
tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 1.48kB/s]
vocab.json: 899k/? [00:00<00:00, 12.3MB/s]
merges.txt: 456k/? [00:00<00:00, 22.3MB/s]
Device set to use cuda:0
Your min_length=56 must be inferior than your max_length=20.
/usr/local/lib/python3.11/dist-packages/transformers/generation/utils.py:1642: UserWarning: Unfeasible length constraints: `min_length` (56) is larger
  warnings.warn(
[{'summary_text': ' Hugging Face is a pioneering open-source AI platform that empowers developers, researchers'}]
```

Questioner

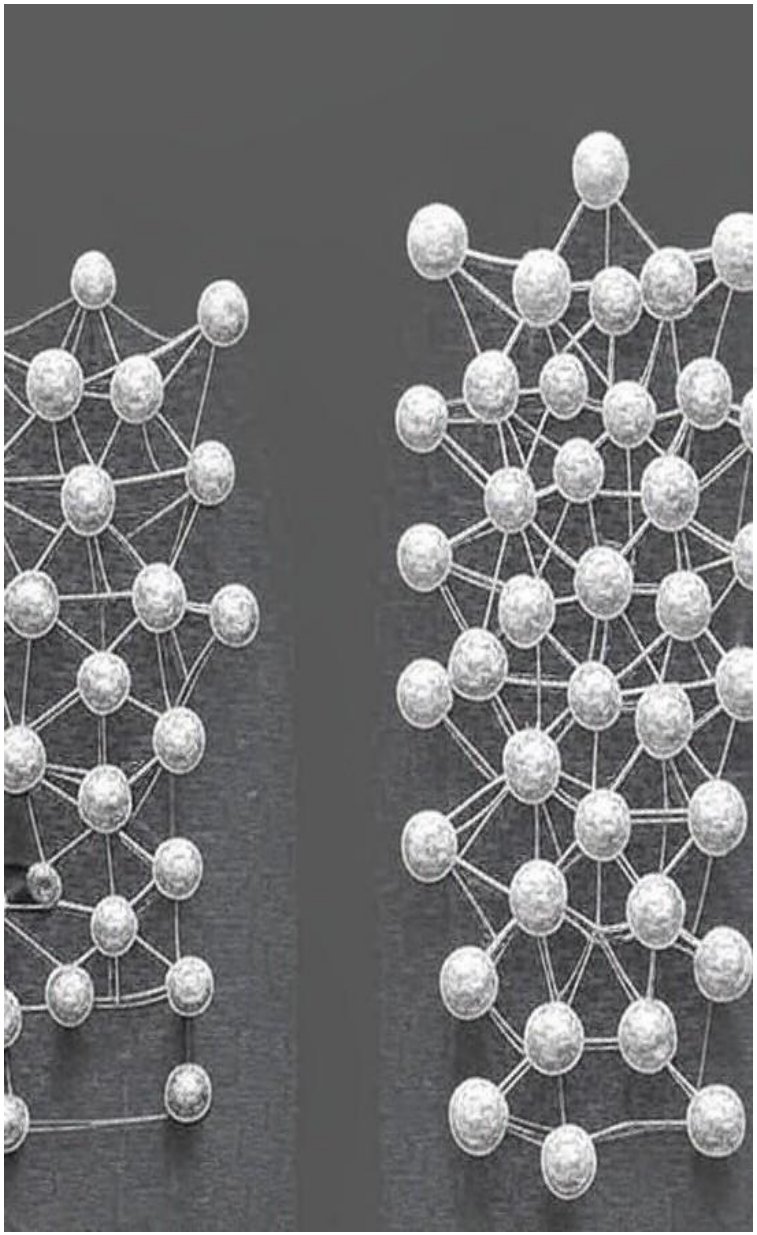


What is a Model?

- A model in machine learning is like a brain made of math.
- Input: Translate 'hello' to French
- Output: 'bonjour'

Some Definitions

- **Foundation Models** – Large pre-trained models (e.g., GPT, PaLM, LLaMA) designed for broad general-purpose tasks.
- **Large Language Models (LLMs)** – A type of foundation model focused on text-based tasks (e.g., GPT-4, Claude, Gemini).
- **Multimodal Models** – Models that handle multiple data types like text, image, audio, and video (e.g., GPT-4o, Gemini, Claude 3).



What are Parameters of a model?

- Parameters are the parts of the model that it learns during training.
- A small model might have 10 million parameters.
- GPT-3 has 175 billion parameters.
- Why do parameters matter?
  - More parameters = more knowledge and capacity to learn patterns.
  - Think of parameters as 'the memory of the model'.

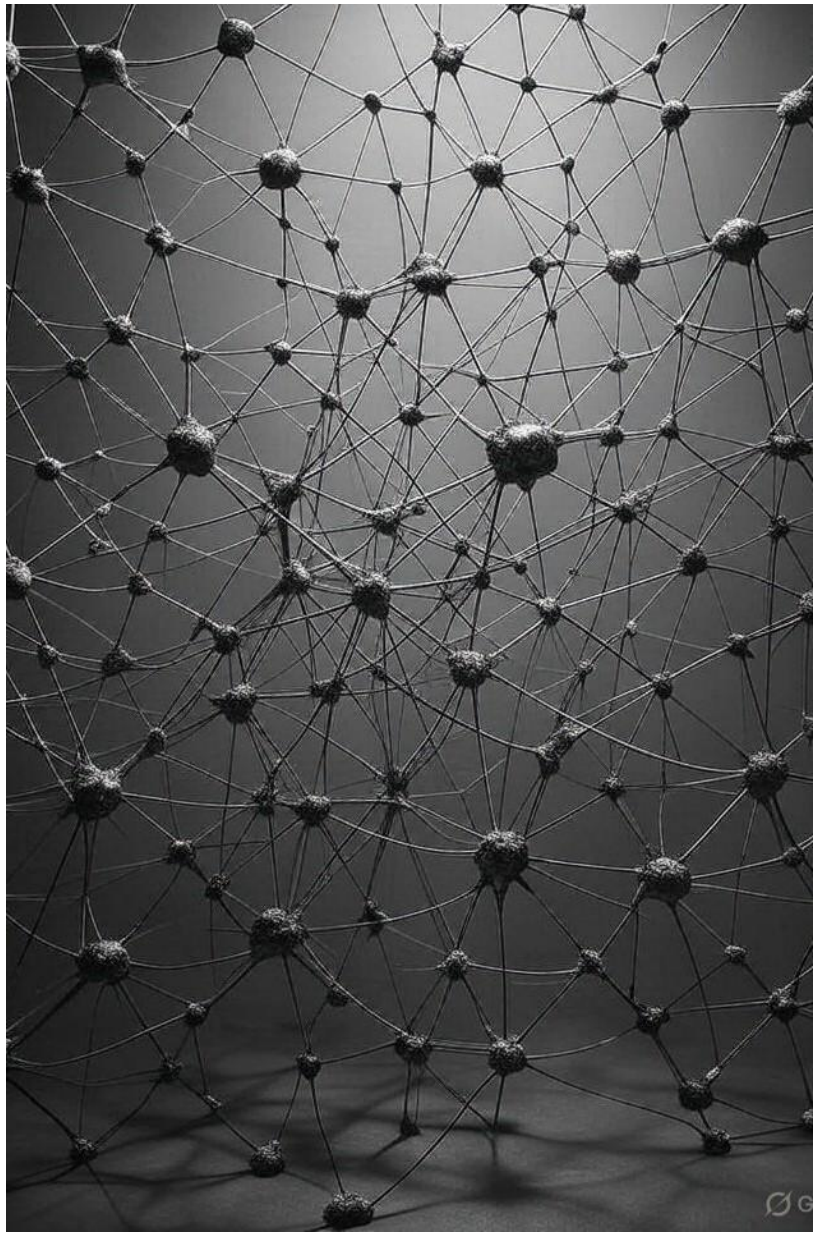
Context Window

- The **context window** is the amount of text a language model can read and understand at once — like its **short-term memory**.
  It's measured in **tokens** (roughly 1 token = ¾ of a word).
- **Why It Matters**
  - Determines how much **previous input** the model can consider when responding
  - Essential for:
    - Long conversations
    - Document summarization
    - Complex reasoning and coding tasks
- **Model Comparisons**
  - **GPT-3.5** can process up to **4,096 tokens** (about 3,000 words)
  - **GPT-4 (Turbo)** extends this to **128,000 tokens** (~96,000 words)
  - **Gemini 1.5** pushes further with **1 million tokens** (~750,000 words)
  - **LLaMA 4** sets a new benchmark with **10 million tokens** (~7.5 million words)



Training a Model

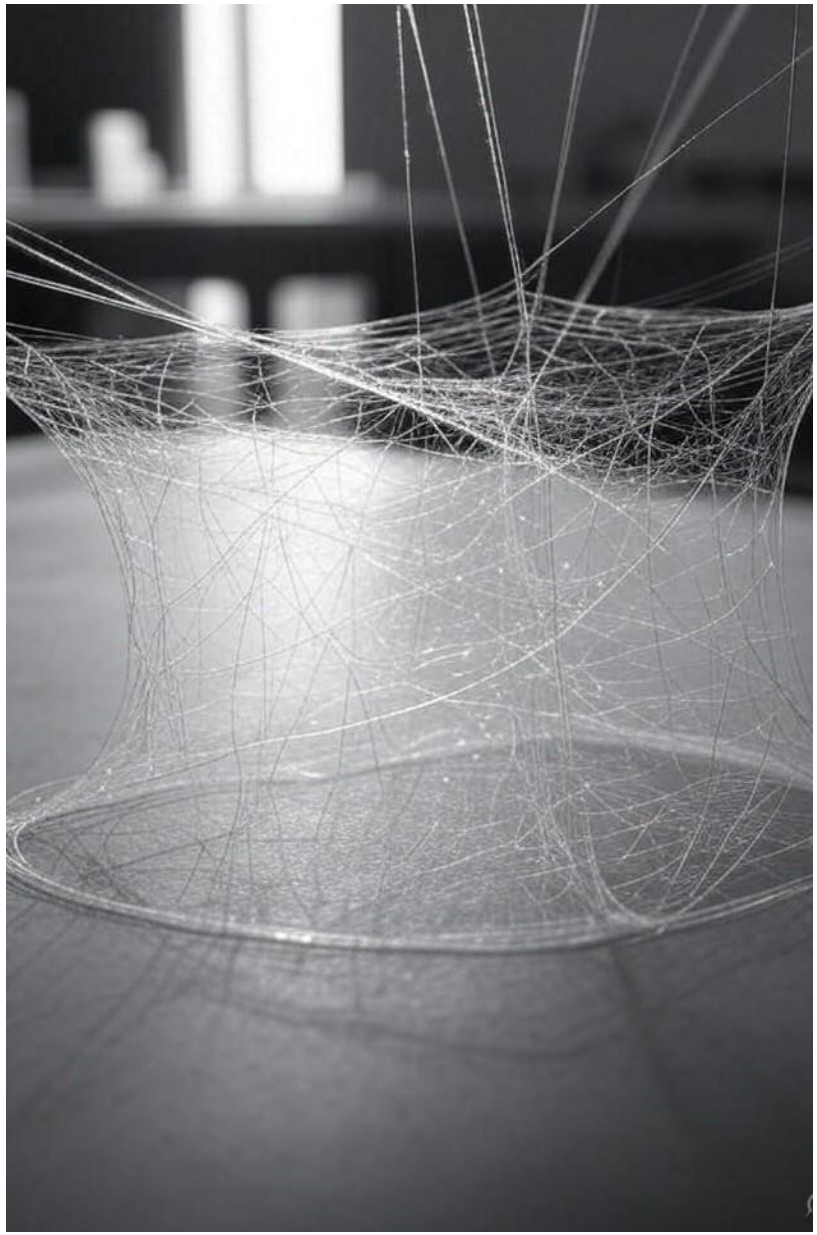
- Training is how a model learns.
  - 1. Give it input (e.g., 'The sky is ___')
  - 2. Give the correct answer (e.g., 'blue')
  - 3. It adjusts parameters to reduce mistakes.
- Training is expensive – it requires:
  - Massive datasets
  - Powerful hardware (like GPUs or TPUs)
  - Days or weeks/months of computation



Using the Model (Inference)

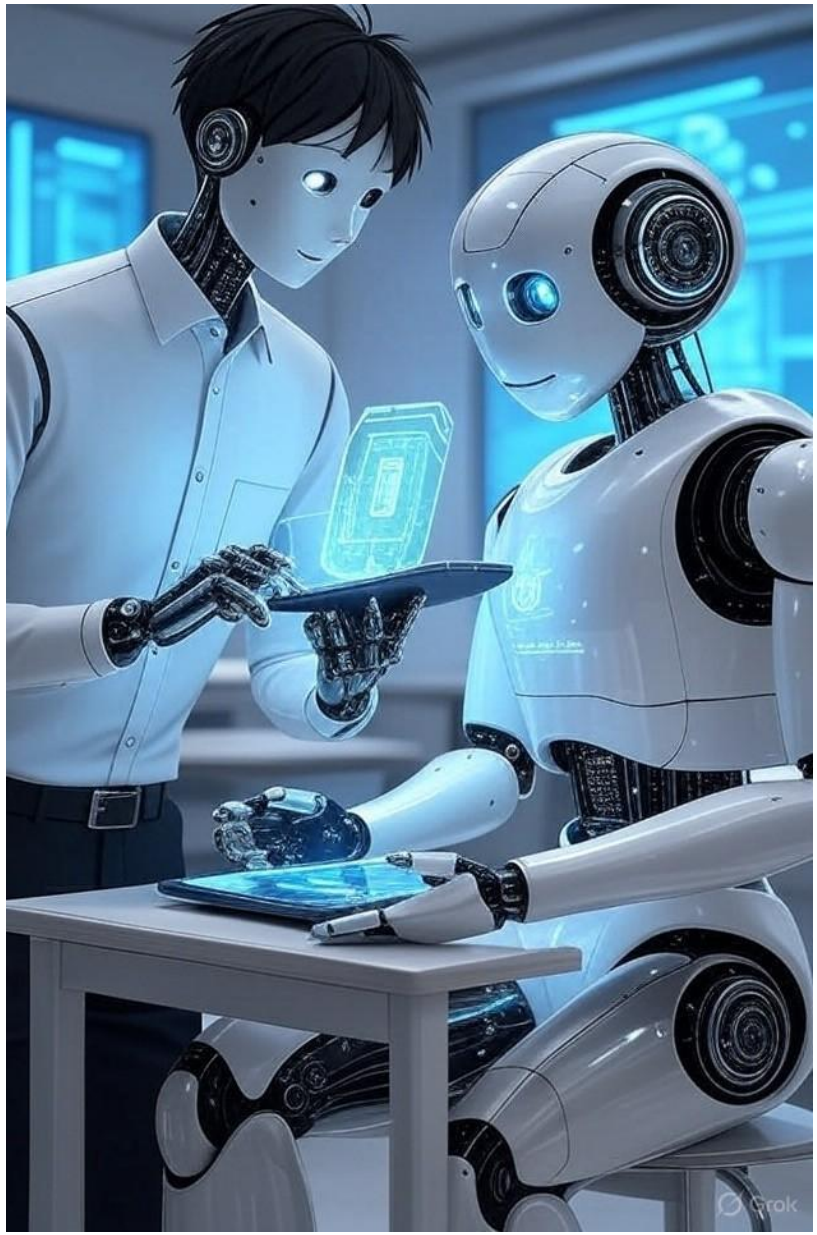
- Inference is using a trained model to get predictions.
- Example: What's the capital of France?
- Output: 'Paris'
- Inference is much faster than training.
- Done on devices like your phone, laptop, or server

  (you need to install the model used)



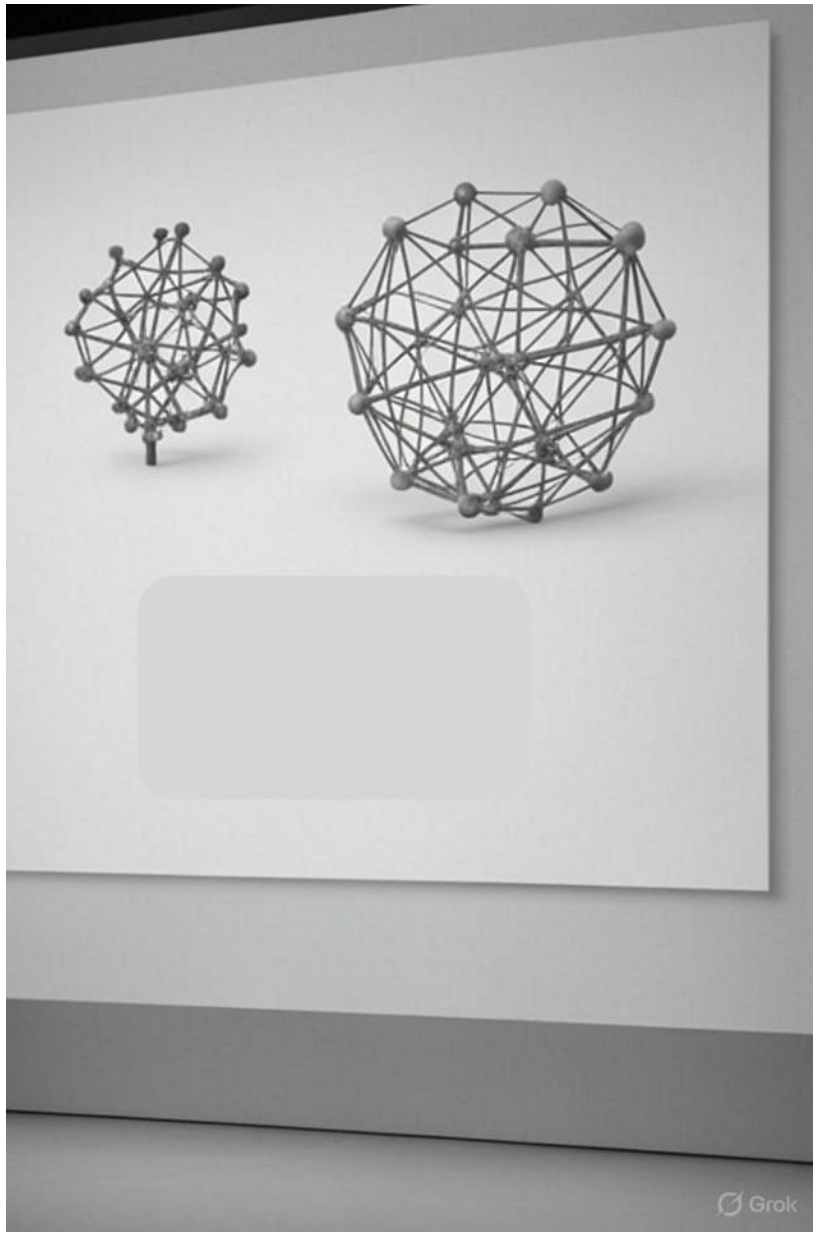
What is Fine Tuning?

- Fine-tuning means taking a pre-trained model and training it on a specific task.
- Why?
  - Better performance on domain-specific data
  - Customize behavior
- Tools: Hugging Face, PEFT, LoRA



What is Distillation?

- Distillation compresses a large model (teacher) into a smaller one (student).
- Why?
  - Faster, lighter, cheaper
  - Edge and mobile deployment
- Example: DistilBERT, TinyBERT

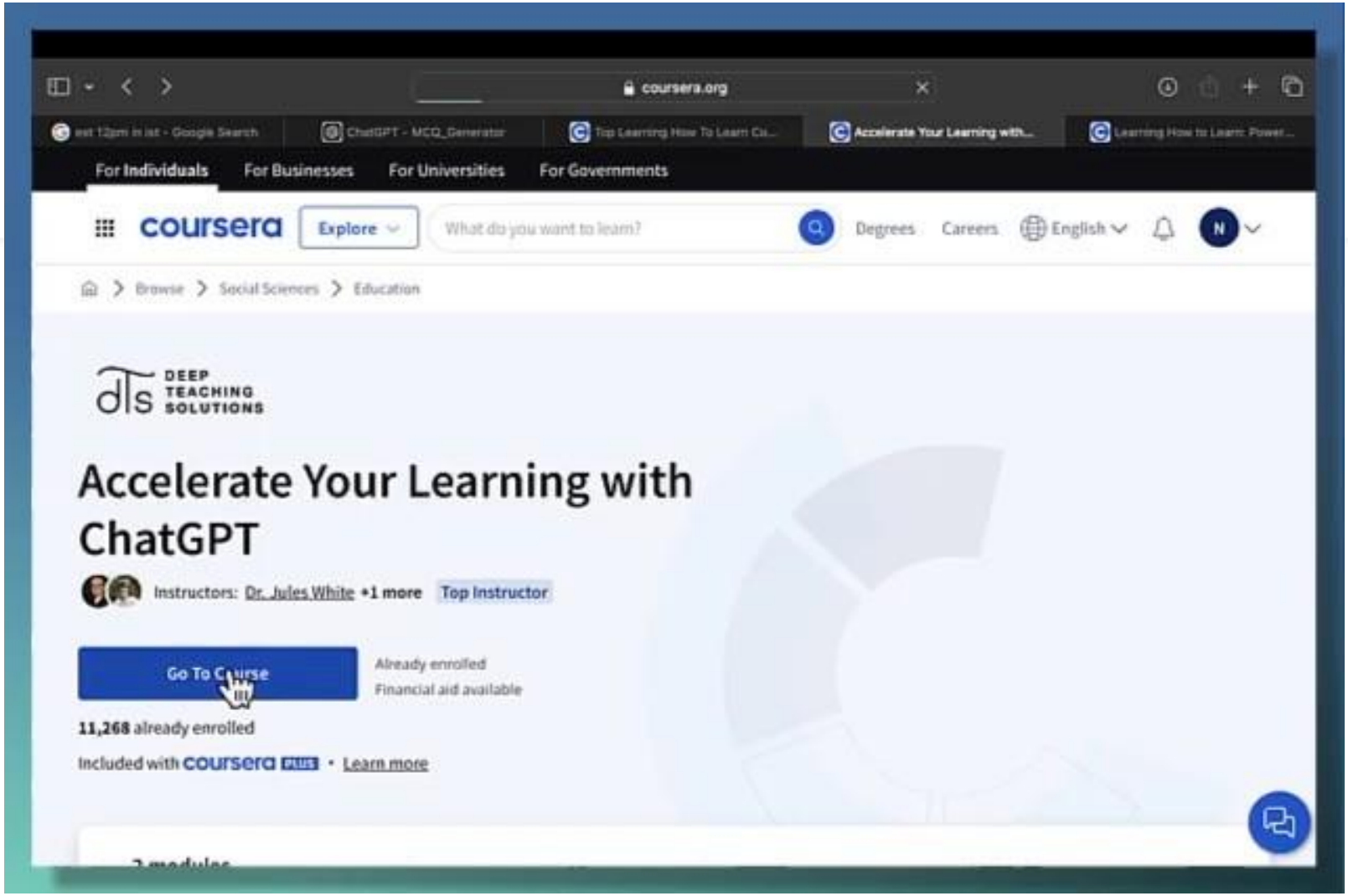


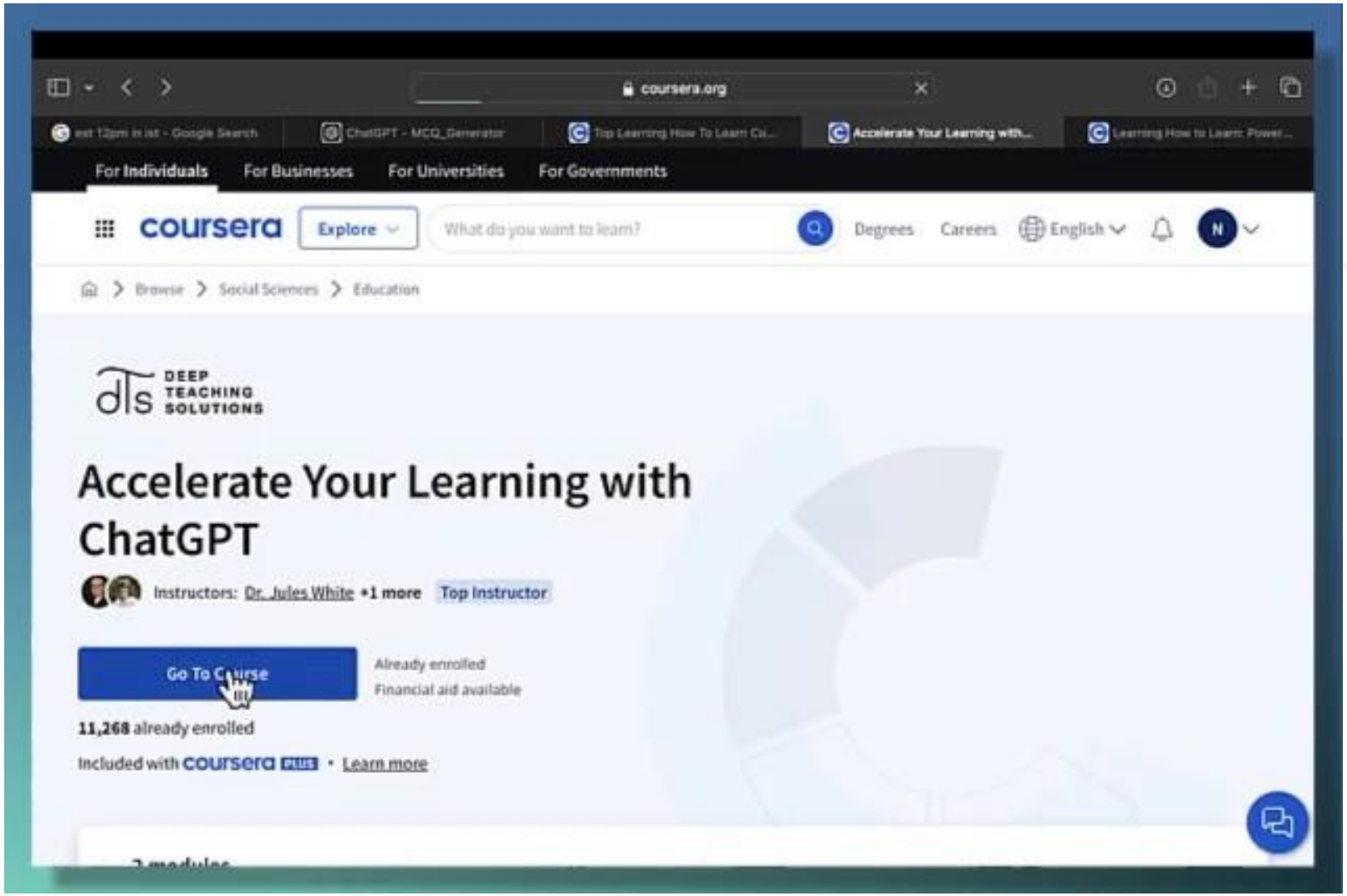
What are Small Language Models?

- SLMs are smaller LLMs optimized for speed and efficiency.
- Examples:
  - TinyLLaMA, Mistral 7B, Gemma 2B, Phi-2
- Use for:
  - Fast inference
  - Low-resource environments

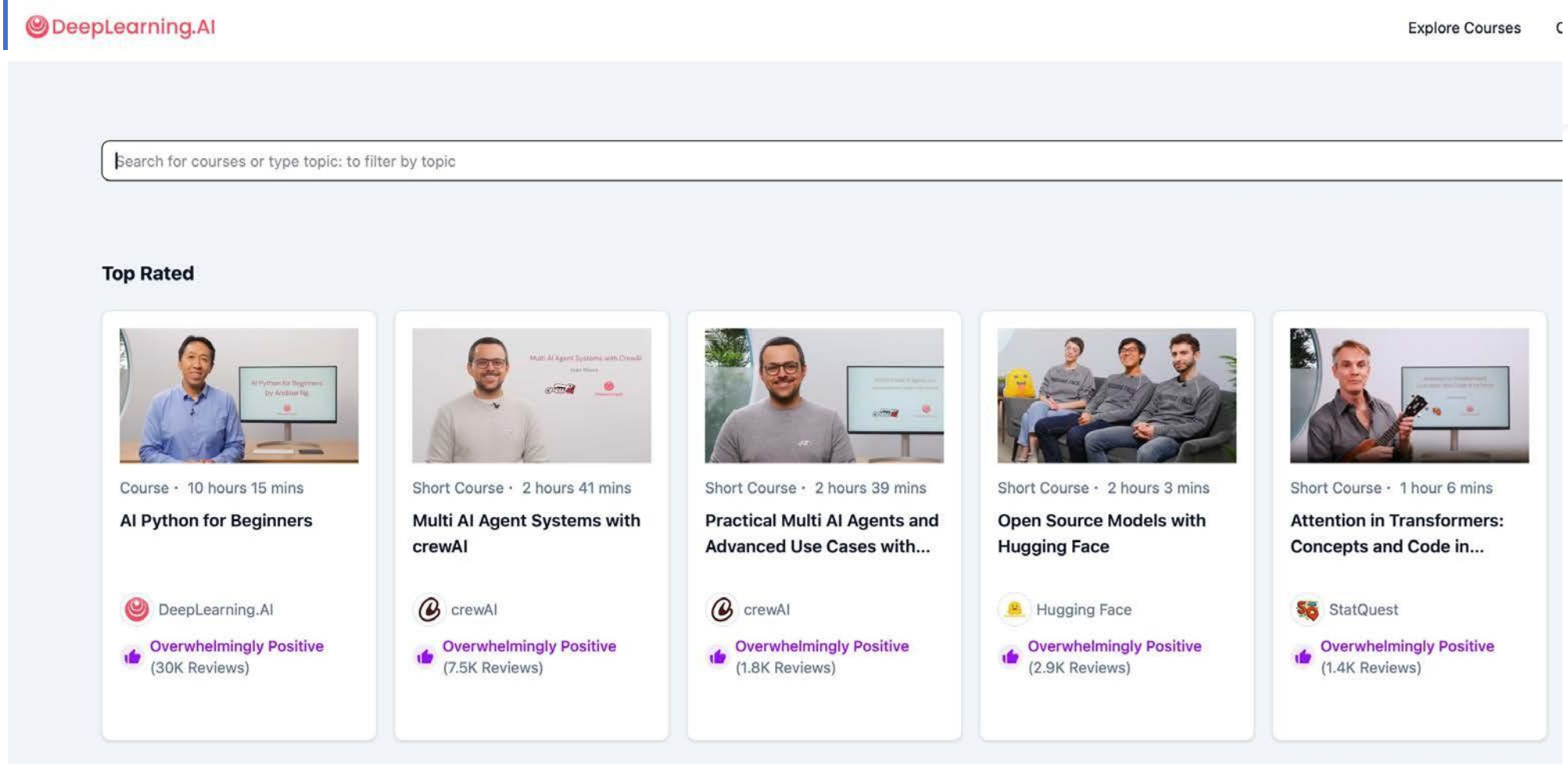
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This course helps you to prepare for the AWS Certified AI Practitioner exam. Review the Bridging to Certification module for details about the exam.

Thank You